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ACADEMIC INTERNSHIP REPORT

**THEME: CONCEPTION AND REALISATION OF AN
APPLICATION TO CONNECT USERS WITH NEARBY
PHARMACIES FOR FAST DRUG AVAILABILITY**

Internship carried out from the 1st of July to the 30th of September 2025 at the
company **REALIZE** as part of my pursuit to obtain a

Higher Technician Diploma (HTD) in Computer Science

Option: Software Engineering

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ACADEMIC YEAR:
2024-2025



CONCEPTION AND REALIZATION OF AN APPLICATION TO
CONNECT USERS WITH NEARBY PHARMACIES FOR FAST DRUG
AVAILABILITY



DEDICATION

THIS BOOK IS DEDICATED TO MY FAMILY

NKENGFACK HEBERT
African Institute of Computer Sciences (AICS) Cameroon
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Level II, Software Engineering for 2024-2025 Academic



ACKNOWLEDGEMENTS

This work wouldn't have been a success without the important contributions of a good number of people who took it upon themselves to see this work accomplished. Our sincere gratitude goes to the following:

- The Resident Representative of IAI-Cameroon, **Mr. Armand Claude ABANDA**, for his support, words of encouragement, and the different advice on how to approach situations.
- The chief executive officer of REALIZE, **Mr. NDABOSE DANIEL**, and his staff for trusting us by allowing us to serve as interns in their prestigious enterprise.
- Our professional supervisor, **Mr. MENGOT NSO**, for his availability, advice, and moral support throughout the internship period, which led to the realization of this project.
- Our academic supervisor, **Mr. ASANE DERICK**, for her encouragement, availability, tolerance, advice, and moral support throughout the internship period.
- To our academic teachers, **Mr. AGBOR**, **Mr. MESSIO**, and **Mme TCHOUTOUO**, for their assistance in the realization of this document.
- To our family members for the guidance, love, moral and financial support throughout this period.
- To all our classmates and friends near and far for their continuous support throughout the drafting of this document.

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GLOSSARY

- **2TUP** : Two-Track Unified Process
- **AICS**: African Institute of Computer Sciences
- **AI**: Artificial Intelligence
- **API**: Application Programming Interface
- **CD-ROM**: Compact Disc Read-Only Memory
- **CSS**: Cascading Style Sheets
- **HTML**: Hypertext Markup Language
- **IAI** : Institut Africain de L'Informatique
- **ICT** : Information and Communication Technologies
- **MERISE** : Méthode d'Étude et de Réalisation Informatique pour les Systèmes d'Entreprise
- **MOMO**: Mobile Money
- **MTN**: Mobile Telecommunications Network
- **OM** : Orange Money
- **TIC** : Technologies de l'Information et de la Communication
- **UI**: User Interface
- **UML**: Unified Modeling Language

ABSTRACT

In today's digital era, timely access to essential medicines remains a major challenge, especially in communities where healthcare infrastructure is limited. Traditional methods of locating pharmacies and confirming the availability of drugs are often slow, unreliable, and inconvenient. To address this gap, this project focuses on the CONCEPTION AND REALISATION OF A MOBILE APPLICATION TO CONNECT USERS WITH NEARBY PHARMACIES FOR FAST DRUG AVAILABILITY. The application utilizes GPS technology to accurately identify the user's location and provide a real-time list of nearby pharmacies, along with information on requested medications. It integrates multiple user roles, allowing customers to search and send availability requests while enabling pharmacies to respond promptly and update their inventories directly. By employing modern mobile development frameworks and secure cloud-based data management, the system ensures real-time communication, reliability, and ease of use. This innovation aims to minimize delays in drug access, enhance healthcare service delivery, and contribute to improved patient outcomes through the seamless connection between users and pharmacies.

KEYWORDS

- ❖ GPS technology
- ❖ Drug availability
- ❖ Mobile Application
- ❖ Healthcare accessibility

RESUME

À l'ère numérique actuelle, l'accès rapide aux médicaments essentiels demeure un défi majeur, en particulier dans les communautés où les infrastructures de santé sont limitées. Les méthodes traditionnelles de localisation des pharmacies et de vérification de la disponibilité des médicaments sont souvent lentes, peu fiables et contraignantes. Pour combler cette lacune, ce projet porte sur la CONCEPTION ET RÉALISATION D'UNE APPLICATION MOBILE POUR CONNECTER LES UTILISATEURS AUX PHARMACIES PROCHES EN VUE D'UNE DISPONIBILITÉ RAPIDE DES MÉDICAMENTS. L'application utilise la technologie GPS afin d'identifier avec précision la position de l'utilisateur et de fournir une liste en temps réel des pharmacies à proximité, accompagnée des informations relatives aux médicaments demandés. Elle intègre plusieurs rôles utilisateurs, permettant aux clients de rechercher et d'envoyer des requêtes de disponibilité, tandis que les pharmaciens peuvent répondre rapidement et mettre à jour directement leurs inventaires. En s'appuyant sur des Framework modernes de développement mobile et une gestion sécurisée des données via le cloud, le système garantit une communication en temps réel, la fiabilité et la simplicité d'utilisation. Cette innovation vise à réduire les délais d'accès aux médicaments, améliorer la prestation des services de santé et contribuer à de meilleurs résultats pour les patients grâce à une connexion fluide entre utilisateurs et pharmacies.

GENERAL INTRODUCTION

In today's digitally connected world, mobile applications have become vital tools for enhancing access to a variety of services, including healthcare. Recognizing the need for a seamless and efficient connection between users and pharmacies, particularly in areas where healthcare infrastructure is limited, this project, titled CONCEPTION AND REALISATION OF A MOBILE APPLICATION TO CONNECT USERS WITH NEARBY PHARMACIES FOR FAST DRUG AVAILABILITY, seeks to address this pressing challenge. The aim is to provide quick and reliable access to essential medications by leveraging location-based technology, thereby reducing the time spent searching for pharmacies and ensuring timely access to drugs.

The development of this project followed a structured approach, beginning with an analysis of existing methods used by individuals to locate nearby pharmacies and the challenges they face, such as limited real-time information and inefficient communication. This was followed by gathering detailed requirements from potential users, including both customers and pharmacy owners, to ensure the application addresses their specific needs. Using standardized modeling tools, we created comprehensive design diagrams to translate these requirements into a coherent system architecture. The implementation phase involved selecting suitable mobile development frameworks and secure cloud-based database solutions, followed by thorough testing to ensure usability, reliability, and responsiveness.

Ultimately, this project contributes to improving healthcare delivery by offering a user-friendly, efficient, and accessible digital solution that connects communities with nearby pharmacies in real time, ensuring that essential medications are readily available when needed.

KEYWORDS:

- ❖ Mobile Application



PART ONE: INSERTION PHASE



Preamble

The inserted document forms part of my internship report, which details the structure and objectives of the organization where I completed my internship. It presents an overview of my internship environment and delineates the different stages I underwent to successfully assimilate into the company

Content:

INTRODUCTION

- I. WELCOMING AND INTERGRATION
- II. GENERAL PRESENTATION OF REALIZE
- III. HARDWARE AND SOFTWARE RESOURCES
- IV. BRIEF PRESENTATION OF THE PROJECT

CONCLUSION

INTRODUCTION

Professional integration is a process that enables individuals or groups to enter the job market under conditions that enhance their chances of securing employment. During our academic internship, we had the opportunity to be intern at REALIZE for three months, with the integration phase spanning two weeks. This section of the internship report outlines the structure and operations of the organization, the experience of the student during the reception period, and the research topic pursued during this time.

I. WELCOMING AND INTEGRATION

The welcoming and integration event took place on the 1st of July at 8:15 am. Upon arrival, we were greeted by Mr. Mengot, the senior developer, who delivered an inspiring welcoming speech. Having interned at REALIZE last year, this moment felt familiar, though it was refreshing to reconnect with the team. We proceeded with a presentation and a meeting focused on discussing the challenges we faced during the coding test, which was a prerequisite for our internship at the company. The meeting also included an overview of the enterprise's structure, its internal policies, and the specific workstations assigned to each intern. Given my previous experience, I was able to quickly reorient myself within the company's environment. Mr. Tanue concluded by encouraging us to strive for excellence and offered his support for any issues we might encounter.

After settling into our designated workstations, we were promptly assigned our first task for the day. During the first week, we conducted a verification of concepts using fundamental HTML and CSS to ensure everyone was on an even playing field before progressing. We then engaged in discussions about project ideas, where we were encouraged to propose our own. My previous experience made it easier to contribute and refine these ideas. Subsequently, we were tasked with researching these ideas, as the enterprise valued our initiative and did not impose any themes on us.

In the second week, we focused on JavaScript concept verification exercises to assess our understanding and reasoning skills. We held a detailed discussion with our professional supervisor about the specification book and its various components, where he outlined what was expected of us in each segment. We also continued developing our project ideas into a concrete plan, benefiting from my previous familiarity with the company's expectations and processes.

II. GENERAL PRESENTATION OF REALIZE

i. Background

Realize is a Cameroon based non-governmental tech start-up institute founded by NDELOKAKEH Daniel in 2022 which proposes IT solutions and empowers the use of new technologies in Cameroon

ii. Missions, Visions and Activities of Realize

1. Missions:

The mission of Realize mainly relates to the empowerment and the perpetuation of Technology use. These missions include:

- Design and realize mobile-oriented software for companies and individuals;
 - Assist newborn companies for quick growth using computer sciences;
 - Provide training and certifications to improve qualified human resources in many fields of study;
- Take an active part in the sustainable development of the world through innovative solutions and virtual reality.

2. Vision:

At Realize, we believe that the true potential of every organization and individual can be unlocked through the harmonious fusion of analysis, development, realization, and data analytics. Our vision propels us forward, guiding us to make a lasting impact on the digital landscape and shape a future where innovation knows no bounds."

3. Activities:

The activities of Realize range from computer sciences, engineering, and training. We can outline the following:

- Conception realization, and hosting of websites;
- Software development and maintenance;
- Training in Software related fields;
- Conception and realization of multimedia;
- IT consulting and innovation
- IT support



1. Localization



Figure 1 Realise localization

III. ORGANIZATION OF REALIZE

A. Administrative organization

Realize is administratively organized as follows:

a. The General management

This is the highest level of the company, which takes care of the following:

- Proper functioning of each department.
- Define project strategies.
- Provide leadership and guidance to the company's employees.
- Makes critical decisions that could affect the company's operations or reputation
- Severe as a point of contact for key stakeholders such as investors etc.

b. Human resource department

This department is in charge of the following:

- Recruitment and Hiring
- Acts as a liaison between employees and management;
- Manages employee compensation and benefits programs;
- Develops and implements company policies and procedures, ensuring compliance with employment laws and regulations.
- Manages administrative tasks related to employee records, data management, and HR systems

c. Communication Department

This department is in charge of the following:

- Handles public relations activities, which involve managing the company's reputation and image;



- Responsible for crafting and delivering messages to external stakeholders such as customers etc.;
- It fosters effective communication within the company;
- They create and manage content across different platforms and channels;

d. Department of Financial Affairs

This department is in charge of the following:

- Responsible for developing and managing the company's financial plans and budgets;
- Prepare and presents accurate and timely financial reports to management, stakeholders, and regulatory authorities;
- Responsible for managing cost and expenses within the organization;
- Ensures compliance with tax laws and regulations. Etc.

e. Technical department

This department is in charge of the following:

- Responsible for managing the company's technological infrastructure, including networks, hardware;
- Play a role in managing and maintaining the company's data;
- Responsible for implementing and maintaining cyber security measures to protect the company's digital assets from potential threats;
- Interacts with technology vendors and manages relationships with external service providers. Etc.

f. Software Engineering Department

This department is in charge of the following:

- Primarily responsible for developing software applications and systems;
- Responsible for maintaining and supporting software application throughout their lifecycle;
- Evaluation and realization of projects etc.

B. FUNCTIONAL ORGANIZATION OF REALIZE

The functional branch of Realize is organized as follows

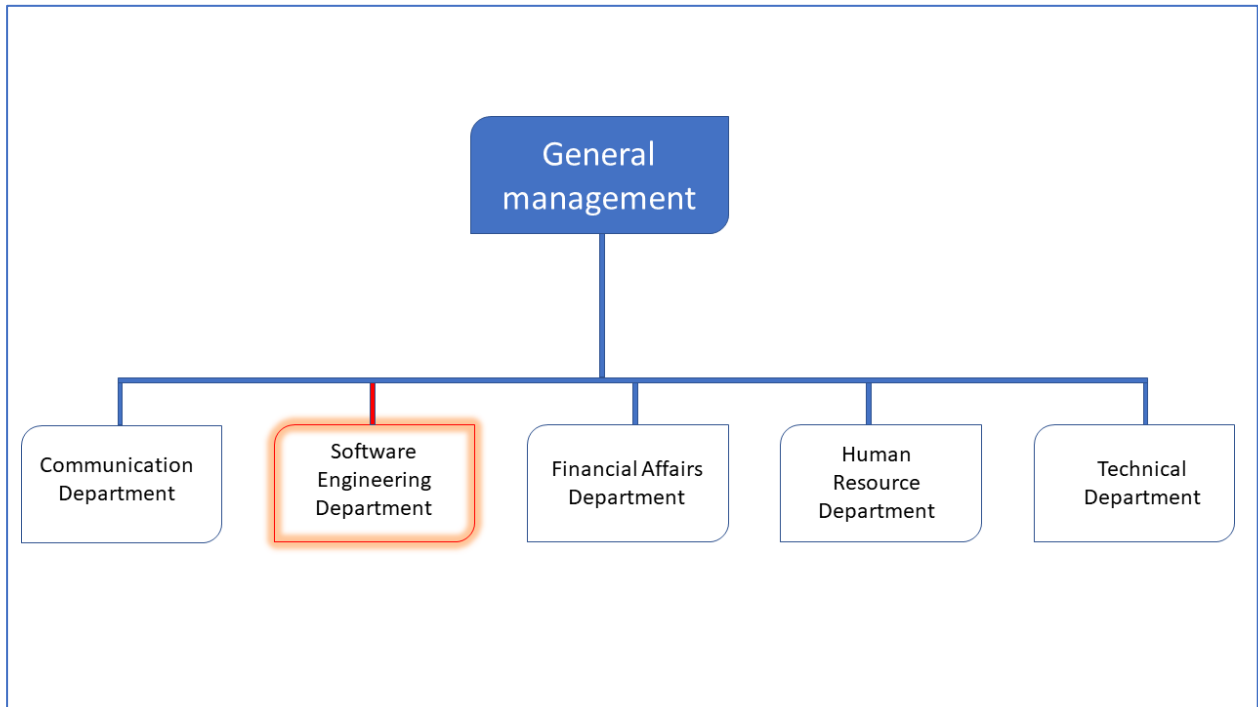


Figure 2 functional branch of realize

IV. HARDWARE AND SOFTWARE RESOURCES

A. HARDWARE RESOURCES

Table 1 hardware resource

No	DESIGNATION	QUANTITY	CHARECTERISTICS	OBSERVATION
1	MacBook Pro	1	APPLE	Good
2	Modem	1	CAMTEL	Good
3	LCD Screen	1	HP	Excellent
4	Training Equipment and office furniture	/	/	/

B. SOFTWARE RESOURCES

Table 2 software resources

DESIGNATION	SOFTWARE
Operating System	Windows 10, Mac OS, parrot Linux
Design tools	Photoshop
Integrated development environment (IDE)	Visual studio code
Text editor	Sublime text, notepad++
Database management system (DBMS)	MongoDB, PostgreSQL, MySQL
Web browser	Google chrome, Microsoft edge
Document editor	Microsoft office word
Presentation	Microsoft office PowerPoint

V. BRIEF PRESENTATION OF PROJECT

In the framework of my internship, I am assigned to develop a mobile application titled "CONCEPTION AND REALIZATION OF A MOBILE APPLICATION TO CONNECT USERS WITH NEARBY PHARMACIES FOR FAST DRUG AVAILABILITY." This project involves creating a digital platform that enables users to locate nearby pharmacies and check the real-time availability of medications. The application aims to simplify the process of finding and obtaining drugs, particularly in urgent situations, by providing a user-friendly interface and reliable information. Using modern technologies such as React Native, Node.js, and MongoDB, and applying system analysis methodologies like UML and the 2TUP process, the application will ensure efficient development and a robust product. The primary objective is to enhance healthcare accessibility and support communities by leveraging technology.

CONCLUSION

Our two-week integration period at REALIZE provided an immersive introduction to an exceptional professional environment that champions excellence, dedication, and collaborative spirit—qualities highly sought after by aspiring technology professionals. During this formative period, we gained comprehensive insights into the company's mission-driven approach and diverse technological initiatives, while establishing meaningful connections with both permanent staff and fellow interns. This orientation culminated in the assignment of our core project theme: "Conception and Realization of a Mobile Application to Connect Users with Nearby Pharmacies for Fast Drug Availability."



PART TWO: TECHNICAL PHASE

Preamble

This chapter will present and elucidate our central theme, which is the primary emphasis and objective of our project. We shall also expound on the predicament that prompted us to devise our proposed solution, and detail the characteristics and operations of the software we have planned to construct.

Content:

INTRODUCTION

- THEME PRESENTATION
- STUDY OF THE EXISTING SYSTEM
- LIMITATION OF THE EXISTING SYSTEM
- PROBLEMATIC
- PROPOSED SOLUTION

CONCLUSION



FILE ONE: EXISTING SYSTEM

Preamble

This chapter will present and elucidate our central theme, which is the primary emphasis and objective of our project. We shall also expound on the predicament that prompted us to devise our proposed solution, and detail the characteristics and operations of the software we have planned to construct.

Content:

INTRODUCTION

- I. PRESENTATION OF THE THEME
- II. STUDY OF THE EXISTING SYSTEM
- III. LIMITATIONS OF THE EXISTING SYSTEM
- IV. PROBLEMATIC
- V. PROPOSED SOLUTIONS

CONCLUSION

INTRODUCTION

As engineers, our primary role in society is to solve problems, making life easier for everyone. To develop effective solutions, we must first ask ourselves key questions: Why does the problem exist, and how is it currently being addressed? Often, we discover that an existing system is already in place to tackle the issue, regardless of its nature. Our task then becomes either to develop a new system or to enhance the existing one. Regardless of the approach, a deep understanding of the current system's functionality is crucial to ensure that our proposed solution truly makes a difference. System development can be broadly divided into two essential components: system analysis and system design. During the analysis phase, we evaluate the current system and propose a new one, determining whether it is desirable or if the existing system simply needs improvement. The ultimate goal of our proposed solution is to raise awareness, provide care, and offer protection to people worldwide.

I. PRESENTATION OF THE THEME

The theme of this project is the **CONCEPTION AND REALISATION OF AN APPLICATION TO CONNECT USERS WITH NEARBY PHARMACIES FOR FAST DRUG AVAILABILITY**, developed to address challenges in accessing timely medication in Cameroon. This mobile application seeks to empower patients and families by providing an accessible and reliable system that simplifies the process of locating essential drugs in their vicinity.

The app enables users to send availability requests for specific medications to nearby pharmacies and instantly receive responses indicating which pharmacies currently stock the requested drugs. This reduces time spent searching for medication, prevents treatment delays, and supports better health outcomes. By integrating mapping and notification services, the platform allows users to easily identify the closest pharmacy with the required drug and navigate directly to it.

The system caters to both users and pharmacies: users benefit from faster, more informed access to essential medicines, while pharmacies can expand their visibility, build trust, and improve customer engagement. To ensure credibility, the app incorporates a verification process for pharmacies, guaranteeing that only accredited and reliable providers are listed.

Healthcare providers and pharmacists can integrate their services within the platform, offering not only drug availability information but also additional services such as consultations, promotions, or prescription management. The system design follows rigorous methodologies, including UML modeling and iterative development processes such as 2TUP, ensuring a secure, scalable, and user-friendly solution.

Ultimately, this project contributes to improving healthcare delivery by bridging the gap between patients and pharmacies, reducing barriers to drug access, and promoting efficient, patient-centered healthcare in Cameroon.

II. STUDY OF THE EXISTING SYSTEM

In the current system, access to essential drugs in Cameroon is often informal, fragmented, and unreliable. Many patients and families depend on personal networks, word-of-mouth, or repeated visits to multiple pharmacies to determine whether a needed medication is available. While these methods may seem accessible, they lack the accuracy, speed, and reliability required for effective healthcare, often resulting in delays in treatment, increased stress, and, in some cases, worsening health outcomes. For those seeking structured support, the situation remains challenging. Only a limited number of pharmacies provide timely drug availability information, and very few maintain an effective online presence. These services are generally concentrated in urban centers, leaving many patients in rural and semi-urban areas without convenient access to reliable information. The absence of a centralized digital platform forces patients to manually contact or visit multiple pharmacies, which is time-consuming, inefficient, and unstandardized.

Additionally, the current drug access system suffers from a lack of transparency and trust. Without integrated verification processes or up-to-date databases, users find it difficult to assess the credibility of pharmacies or the accuracy of drug availability claims. This gap is further compounded by the limited use of digital tools by many pharmacies, making reliable information scarce and inconsistently distributed.

This fragmented and inaccessible landscape highlights the urgent need for a unified, user-friendly, and trustworthy platform. The proposed mobile application, developed under the theme of “Conception and Realization of an Application to Connect Users with Nearby Pharmacies for Fast Drug Availability,” directly addresses these challenges. It enables patients to request real-time information on drug availability, receive instant notifications from verified pharmacies, and navigate easily to the nearest pharmacy stocking the needed medication.

Built using rigorous design methodologies such as UML modeling and the 2TUP process, the platform ensures reliability, security, and scalability. By integrating mapping services, notification systems, and pharmacy verification processes, the application creates a standardized, transparent, and accessible solution. This not only improves access to essential medicines but also strengthens trust between pharmacies and patients, ultimately enhancing healthcare delivery in Cameroon

III. LIMITATIONS OF THE EXISTING SYSTEM

Table 3 Limitations of the existing system

LIMITATIONS	CONSEQUENCES	PROPOSED SOLUTION
Lack of Real-Time information	Users don't know which pharmacy has the medication they need; leads to multiple physical visits.	Real-time drug availability updates from pharmacies through the mobile app.
Time-Consuming Search Process	Patients waste time visiting or calling many pharmacies, especially in emergencies.	Users can send one notification to all nearby pharmacies and receive quick responses in-app
Fragmented And Outdated Information	Patients waste time visiting or calling many pharmacies, especially in emergencies	Centralized and regularly updated digital platform listing verified pharmacies and their stock status.
Fragmented And Outdated Information	Printed directories and informal sources are often inaccurate or obsolete.	Centralized and regularly updated digital platform listing verified pharmacies and their stock status.
Limited Accessibility	Remote users lack nearby pharmacies or access to reliable information sources.	GPS-based search shows nearest pharmacies with available drugs, even outside the user's immediate area.



IV. PROBLEMATIC

In Cameroon, limited access to timely drug information, lack of centralized pharmacy services, and the absence of digital tools for medication tracking contribute to delays in treatment and poor health outcomes. Patients often waste valuable time searching for drugs, which can worsen illnesses and increase healthcare costs. These challenges raise the key question: HOW CAN WE IMPROVE ACCESS TO MEDICINES IN CAMEROON THROUGH A RELIABLE AND USER-FRIENDLY DIGITAL PLATFORM?

V. PROPOSED SOLUTION

To address the identified challenges, we propose developing a comprehensive mobile application to connect users with nearby pharmacies for fast drug availability. This platform, built using technologies such as React Native, Node.js, and MongoDB, and designed following UML and the 2TUP process, offers a centralized solution for improving access to medicines in Cameroon. It aims to enhance efficiency, transparency, and user experience for both patients and pharmacies.

- **Real-Time Drug Information:** Users can send requests and instantly check the availability of drugs in nearby pharmacies, reducing delays in treatment.
- **Accessibility and Convenience:** Integrated mapping and notification services guide users to the nearest pharmacy stocking their medication.
- **Varied Services:** Pharmacies can provide additional services such as promotions, prescription validation, or consultation, meeting diverse healthcare needs.
- **Transparency and Trust:** Verified pharmacies list their services, supported by user reviews and ratings to ensure credibility.
- **Reliability:** Only accredited and trustworthy pharmacies are highlighted, guaranteeing dependable information and services.

CONCLUSION

In conclusion, after analyzing the current methods for accessing medications and interacting with pharmacies in Cameroon, we identified key challenges faced by users, including delays, outdated information, and inefficient communication. To address these issues, we proposed a practical and innovative solution through the conception and realization of a mobile application to connect users with nearby pharmacies for fast drug availability. This system is designed to streamline the search for medications, provide real-time updates on drug stock, and enhance accessibility for both users and pharmacy providers. Ultimately, the application aims to modernize the process of obtaining essential medicines and improve healthcare delivery in the digital age

FILE 2 : SPECIFICATION BOOK

Preamble

The specification book serves as a contractual document before the application development process commences. It succinctly outlines the expectations of the project owner and the respective responsibilities of all parties involved. It provides a specific and detailed description of the topic and user requirements, as well as the necessary conditions for project execution. The project's success will depend on careful allocation and management of the various resources available, including physical, financial, and human assets. Therefore, this document outlines the criteria that the developer must adhere to in order to ensure the success of the project.

Content:

INTRODUCTION

- I. CONTEXT AND JUSTIFICATION
- II. OBJECTIVES OF THE PROJECT
- III. EXPRESSION OF NEEDS
- IV. PROJECT PLANNING
- V. ESTIMATED COST OF THE PROJECT
- VI. LIST OF PARTICIPANTS
- VII. PROJECT CONSTRAINTS
- VIII. DELIVERABLES.

CONCLUSION

INTRODUCTION

The report's specification book enables us to provide comprehensive details about our topic, enhancing our understanding and improving its chances of success. To establish the project's scope, we will describe the context of our topic and identify the issues within it that we aim to solve. Once the solution is proposed, we will outline the objectives we have set for ourselves and for the project. Also, we will assess the functional and non-functional requirements of our system before designing a plan to ensure the timely completion of the project and determine the financial resources needed. Subsequently, we will outline the expected project deliverables at the conclusion of the project.

I. CONTEXT AND JUSTIFICATION

a. Context

Reliable and efficient access to medicines is crucial for effective healthcare delivery in Cameroon, where patients often face delays, misinformation, and limited access to pharmacies. Many individuals spend valuable time moving from one pharmacy to another in search of essential drugs, which can result in treatment delays and worsening health conditions.

Currently, the process of checking drug availability is informal, fragmented, and difficult to navigate. Patients rely on word-of-mouth or physical visits, often encountering incomplete or outdated information. This lack of structure leads to wasted time, higher costs, and reduced trust in the healthcare system.

This project aims to overcome these challenges by developing a mobile application that connects users with nearby pharmacies for fast drug availability. The platform empowers users with real-time access to verified pharmacies, integrated mapping and notification services, and transparent information to support quick and informed decisions. By leveraging modern technologies and robust design methodologies, the app ensures a secure, reliable, and user-friendly solution.

The growing adoption of mobile health solutions and the urgent need to improve access to medicines in Cameroon make this project both timely and highly relevant.

b. Justification

Reliable and efficient access to medicines is crucial for effective healthcare delivery in Cameroon, where patients often face delays, misinformation, and limited access to pharmacies. Many individuals spend valuable time moving from one pharmacy to another in search of essential drugs, which can result in treatment delays and worsening health conditions.

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The growing adoption of mobile health solutions and the urgent need to improve access to medicines in Cameroon make this project both timely and highly relevant.

II. OBJECTIVES OF THE PROJECT

A. GENERAL OBJECTIVE

Our general objective is to develop and implement a user-friendly mobile application that improves access to medications by connecting users with nearby pharmacies in real-time, enabling drug availability verification, reducing delays in treatment, and enhancing the overall efficiency of the drug search and availability process in Cameroon.

B. SPECIFIC OBJECTIVES

- Facilitate quick access to drug availability information
- Enhance user experience with real-time notifications and maps
- Ensure transparency in pharmacy services and drug listings
- Reduce unnecessary physical visits to multiple pharmacies
- Guarantee timely access to essential medicines
- Build trust through verified pharmacy profiles
- Support patients with accurate, reliable, and updated information

Project Characterization

Project Name: PharmaConnect

Project Target: Patients in need of medication, families seeking quick access to essential drugs, and pharmacies aiming to expand visibility and service delivery

Technical Specification: Mobile application (React Native) integrated with a backend (Node.js, MongoDB) for:

Real-time drug availability management

Secure authentication and user accounts

Pharmacy verification and listing services

Integrated mapping and notification system

Optional payment/booking for pharmacy services

III. EXPRESSION OF NEEDS

1. Functional Needs

In this context, functional requirements describe what the system or application should do. The modules are as follows;

Patient:

- Allow users to search for nearby pharmacies based on their current location
- Perform Display real-time medication availability from pharmacies
- Enable pharmacies to update their stock and drug availability.
- Facilitate user registration and Login for different roles (client, pharmacist and admin)
- Send notifications to users about drugs availability and requestnotifucation
- Chat Allow users to request drugs and search from various pharmacies
- Rate provide pharmacies with a good dashboard to manage inventory

Health Expert / Medical Doctor:

- Respond to Drug Requests: Pharmacies review and respond to user inquiries about drug availability in real time.
- Provide Recommendations: Pharmacies suggest alternative drugs or dosage options when requested medicines are out of stock.
- Manage Inventory: Pharmacies update and track drug stock levels to ensure accurate availability data.
- Update Drug Information: Pharmacies validate and update details on new arrivals, promotions, or changes in drug availability.
- Publish and Update Health Tips: Admins manage the publication of educational tips, awareness content, and safe drug usage guidelines.

Administrator:

- Register and Approve Health Experts: Admins verify and approve health expert accounts and credentials.



- **Manage User Accounts:** Admins oversee the registration and management of all users, including patients and experts.
- **Validate Crisis Management Content:** Admins review and approve updated crisis management protocols and health tips.

2. Non-Functional Needs

In Performance Requirements

- **Handle Concurrent User Actions:** The system must support multiple users simultaneously sending drug availability requests, browsing pharmacies, or receiving notifications without lag, especially during peak times.
- **Fast Response Times:** Actions such as checking drug availability, receiving confirmation, or navigating to a pharmacy must complete in under 3 seconds to provide a seamless user experience.

Security Requirements

- **Data Encryption:** Encrypt all sensitive data, including user profiles, pharmacy details, and payment information, to protect user interactions and maintain trust.

Scalability Requirements

- **Support Growing User Base:** The system must scale to accommodate increasing numbers of users submitting requests, browsing pharmacies, or making bookings without affecting performance.
- **Horizontal Scaling:** Enable horizontal scaling for both frontend (user app interface) and backend (drug availability service, notifications, and payment processing) to handle growing demand efficiently.

PROJECT PLANNING

a. Chronogram of activities

We began our internship on the 1st of July, till the 30th of September 2025, a total of 3 months. To this, it was an obligation for us to manage our time properly so as to finish on time. This section presents how the work phases were scheduled throughout my internship period.

Table 4 Chronogram of activities

PHASE	OBJECTIVE	OUTPUT	DURATION	PERIOD
INSERTION	Collection of information on the enterprise	Insertion book	2 weeks	08 th July to 19 th July
EXISTING SYSTEM	Study of the existing system	Existing System	1 week	22 nd July to 26 th July
SPECIFICATION BOOK	Specification of the user needs	Specification Book	1 week	29 th July to 02 nd August
ANALYSIS	Capture of needs Use case and textual description Modelling	Analysis Book	2 weeks	05 th August to 16 th August
CONCEPTION	Preliminary conception and Detailed conception	Conception book	2 weeks	19 th August to 30 th August
REALIZATION	Implementation Unitary test Integration Test Development	Realization book	3 weeks	02 nd September to 20 th September
TEST OF FUNCTIONALITIES	Testing of the software and debugging	Test of functionalities	1 week	23 rd September to 27 th September
INSTALLATION AND USER GUIDE	Documenting software	User Guide	1 week	30 th September to 04 th October

b. Gantt Project planning

Figure 3 PharmaConnecte Gantt project planning

IV. ESTIMATED COST OF THE PROJECT

a. Software Resources

For the realization of this project, we will need the following software applications:

Table 5 Software resources for pharmaconnect

RESOURCES	DESIGNATION	USAGE	QUANTITY	UNIT COST (FCFA)
Formatting	MICROSOFT OFFICE PROFESSIONAL 2019	Writing of the report	1	355465
Mobile Testing	Expo Go (Mobile Device)	Testing React Native-based mobile app features	1	Freeware
Code Editor	Visual Studio Code	For writing the code of the application	1	Freemium
Project Planning	Gannt Project	For building a Gantt chart	1	Freemium
UML Analysis	Visual Paradigme	Drawing UML diagrams to map action	1	Community Version
Version Control	Git & GitHub	For tracking and managing changes to software code	1	Free Version
TOTAL 1			6	355465

b. Hardware Resources

This includes the material resources used to realize this project. The table below shows a representation of these different materials:

Table 6 Hardware Resources for Pharmaconnect

RESOURCES	HARDWARE	QUANTITY	UNIT COST (FCFA)
Computer	Dell latitude (Intel Core i7-7600U, 16GB RAM, 512GB SSD)	1	200,000
Network	Local network installation	1	226,000
Removable Disk	USB HP 2.2 FLASH DRIVE V225W - 8 GB	1	4,160
Smartphone	IPhone 12pro 128gb	1	240,000
Total 2		4	440,160

c. Human resources

Table 7 Human resource for PharmaConnect

ROLE	NUMBER OF DAYS	QUANTITY	COST PER DAY	TOTAL PRICE (FCFA)
Project Manager	90	01	30 000	2 700 000
Analyst	21	01	25 000	525 000
UI/UX Designer	30	01	10 000	140 000
Programmer	30	01	15 000	450 000
Tester	14	02	10 000	140 000
TOTAL 3				3 955 000

d. Global Estimation

Table 8 Global estimation

TOTAL 1 (FCFA)	TOTAL 2 (FCFA)	TOTAL 3 (FCFA)	Unexpected charges (10%) {(Total cost) * 10%}	OVERALL TOTAL (FCFA)
355465	440,160	3,955,000	475,063	5,225,688
Five million two hundred twenty-five thousand six hundred eighty-eight.				

V. LIST OF PARTICIPANTS

The table below presents the different persons who took part in the accomplishment of this project. They include:

Table 9 list of participants

NAME	FUNCTION	ROLE
Mr. MENGOT NSO	Follows up interns at the company level	Professional Supervisor
Mr. ASANE DERICK	Follows up student at the academic level	Academic Supervisor
NKEMNGFACK HEBERT	AICS Student Intern	Student at AICS

VI. PROJECT CONSTRAINTS

a) Software constraints

- ❖ **Compatibility:** The platform must be compatible with different devices (smartphones, tablets) to reach a wide audience.
- ❖ **Security:** Robust security measures must be put in place to protect users' personal data and prevent any breach of confidentiality.
- ❖ **User-friendliness:** The user interface must be user-friendly to ensure a positive user experience.

b) Material constraints

- ❖ **Internet availability:** The platform assumes that users have access to the Internet. However, efforts must be made to guarantee an acceptable experience even with limited Internet connections.
- ❖ **Hardware compatibility:** The platform must be designed to run on a variety of devices, taking into account differences in screen size and processing power.

c) Quality constraints

- ❖ **Accuracy of information:** Guides and information provided must be accurate, up-to date and verified, to avoid giving incorrect information to users.
- ❖ **Performance:** The platform must be fast and responsive to avoid excessive loading times and maintain user engagement.
- ❖ **Accessibility:** The platform must be designed to be accessible to people with disabilities, in accordance with web accessibility standards.

d) Time constraints

- ❖ **Deadlines:** The project must be completed within the specified deadlines to meet user needs and stakeholder expectations.
- ❖ **Regular updates:** Future updates and enhancements to the platform must be planned to meet the changing needs of citizens and to maintain long-term relevance.



VII. DELIVERABLES

In this section, we will present the different elements that will be delivered at the end of this project. They include:

- ❖ A full report including:
 - The insertion file,
 - Specifications,
 - The analysis file,
 - The design file,
 - The project file,
 - Installation and user guide.
- ❖ Software Application
- PowerPoint presentation

CONCLUSION

To conclude, the specification book has permitted us to present the general and specific objectives of the project, the project constraints, the functional and nonfunctional needs of the project, lay out the different resources of the project, and finally, the project planning. The end of this document will enable us to easily dive into drafting the analysis document, which will bring out the technical aspects of the project as a whole.



FILE 3 : THE ANALYSIS PHASE

Preamble

The analysis document that adheres to the specifications provides a comprehensive overview of the issue and the proposed solution, supported by diagrams. It outlines the selected analysis method for the project and explains the rationale behind this choice. All of these elements contribute to a clearer understanding of the information system being modeled.

Content:

INTRODUCTION

I. METHODOLOGY

- a. COMPARATIVE STUDY BETWEEN MERISE AND UML
- b. PRESENTATION OF THE 2TUP
- c. CHARACTERISTICS OF THE 2TUP
- d. JUSTIFICATION OF THE CHOICE OF THE ANALYSIS

II. MODELING

- a. USE CASE DIAGRAM
- b. COMMUNICATION DIAGRAM
- c. SEQUENCE DIAGRAM
- d. ACTIVITY DIAGRAM

CONCLUSION

INTRODUCTION

System development can be thought of as having two major components: System analysis and system design, which both help in understanding the details of the existing system or the system to be designed. The analysis and design of information systems have most of the time vocation to allow the creation of databases, which must represent as closely as possible the reality of the field studied thus requiring the use of a design method. This is why our choice will be directed on the UML method as it offers much to developers seeking a user-centered approach and / or a wide scope in design. This part of the report consists of the comparative study of UML and MERISE, unified processes and finally the various diagrams that meet the functional need requirements.

I. METHODOLOGY

Analysis is the fundamental step in the conception of software. It is the basis for the realization of every information system. An information system is an organization of resources (human, software, and hardware), data, and feedback to attain a specific objective. Several methods and languages have been put in place to facilitate the analysis and conception of information systems, which principally have UML and MERISE.

a. COMPARATIVE ANALYSIS BETWEEN MERISE AND UML

i. MERISE

MERISE stands for “Méthode d’Etude et de Réalisation Informatique pour des Systèmes d’Entreprise”. Although it is prescriptive to some extent, MERISE permits the participation of end users and senior management as well as data processing professionals in its decision cycle. MERISE is a method for designing, developing and carrying out IT projects. The goal of this method is to achieve the design of an information system. The MERISE method is based on the separation of data and processing to be carried out in several conceptual and physical models. The essentials of the approach lie in its three cycles: the decision cycle, the life cycle and the abstraction cycle, which cover data and process elements equally. The separation of data and processing ensures longevity in model. Indeed, the arrangement of data does not have to be often overhauled, while treatments are more frequently.

ii. UML

UML (Unified Modelling Language) is a standard notation for the modelling of real-world objects as a first step in developing an object-oriented design methodology. Its notation is derived from and unifies the notations of three object-oriented design and analysis methodologies: Grady Booch's methodology for describing a set of objects and their relationships, James Rumbaugh's Object-Modelling Technique (OMT), Ivar Jacobson's approach which includes a use case methodology. Other ideas also contributed to UML, which was the result of a work effort by Booch, Rumbaugh, Jacobson, and others to combine their ideas, working under the sponsorship of Rational Software. UML captures information about the static and dynamic view of a system. UML 2.5 comprises of 14 diagrams which represent the different views of a system. The 14 diagrams can be subdivided into two, Static or structural and Dynamic diagrams. These diagrams include;



1. STATIC OR STRUCTURAL DIAGRAMS

- Class diagram;
- Object diagram;
- Component diagram;
- Deployment diagram;
- Composite Structure diagram;
- Package diagram;
- Profile Diagram;

2. BEHAVIOURAL OR DYNAMIC DIAGRAMS

- Use case diagrams;
- Activity diagram;
- State machine diagram;
- Sequence diagram;
- Communication diagram;
- Global Interaction diagram;
- Timing Diagram;

These diagrams can be represented on the diagram bellow:

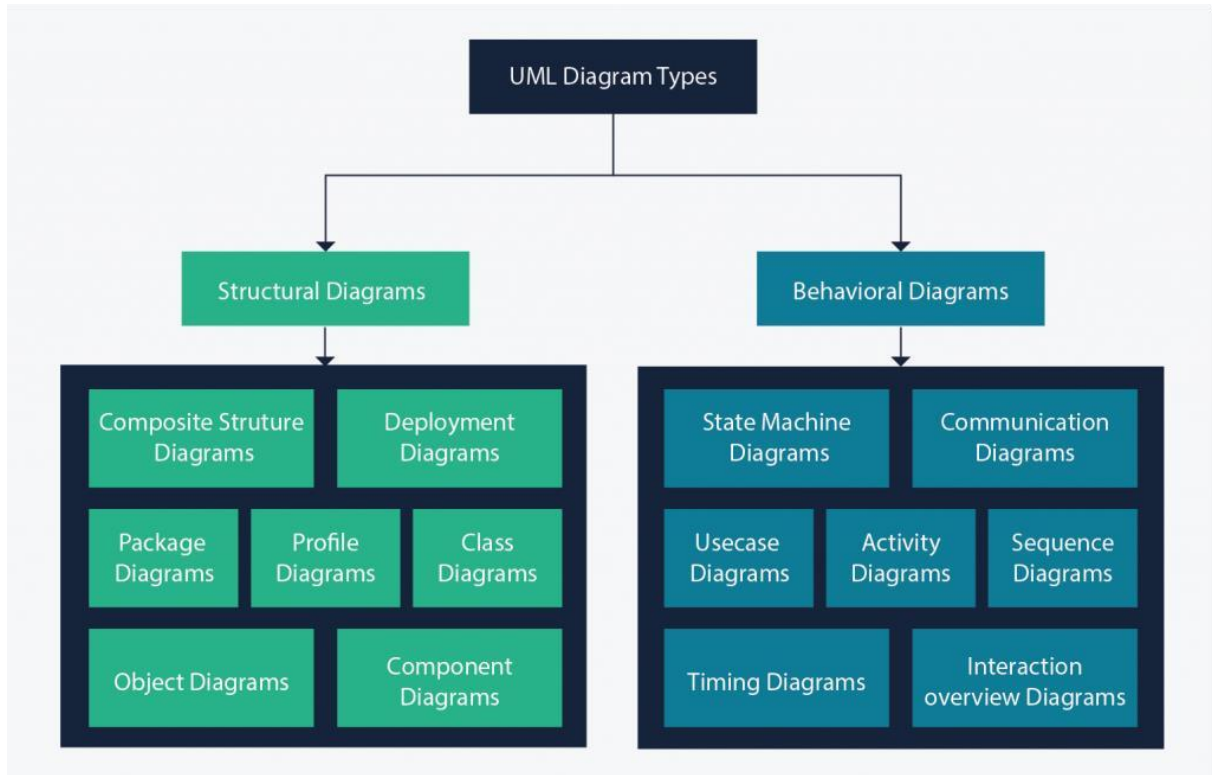


Figure 4 UML Structural and behavioral diagram

Source : <https://images.edrawsoft.com/articles/what-is-uml/uml-diagram-types.png>

Table 10 Comparative analysis between MERISE and UML

Méthode d'Etude et de Realization Informatique pour les Systèmes d'Entreprise	Unified Modeling Language
Merise is a method of analysis.	UML is a modeling language
Merise is a systematic method of analysis and design for information systems, i.e., it uses a system approach.	UML is, however, not a method but rather an object-oriented language that uses the 2TUP approach to model a system
It is based on a procedural approach	Based on the object approach
Merise proposes to consider the real system from two points of view: • The Static view(data) • The Dynamic view(treatments). Which means in Merise, we have a separate study of data and treatments	UML offers a different approach from that of MERISE in that it combines data and process

b. PRESENTATION OF THE 2TUP

2-TUP (Two Track Unified Process) is Y in shape and is a unified process that has as its objective to bring a solution to constraints of functional changes imposed on information systems. It proposes a development cycle that separates the functional aspects from the technical aspects. 2-TUP therefore distinguishes two main branches:

(The functional and technical branch) whose different results are being merged to realize our system.

Any unified process UP (Unified Process) process meets the following characteristics:

- ❖ It is iterative and incremental.
- Some of the advantages of the 2TUP include:
- ❖ It allows the addition of modules.
- ❖ It allows flexibility in the analysis of a project.
- ❖ It is easy to design and understand.
- ❖ It allows to have a preliminary view

Below is the representation of 2-TUP:

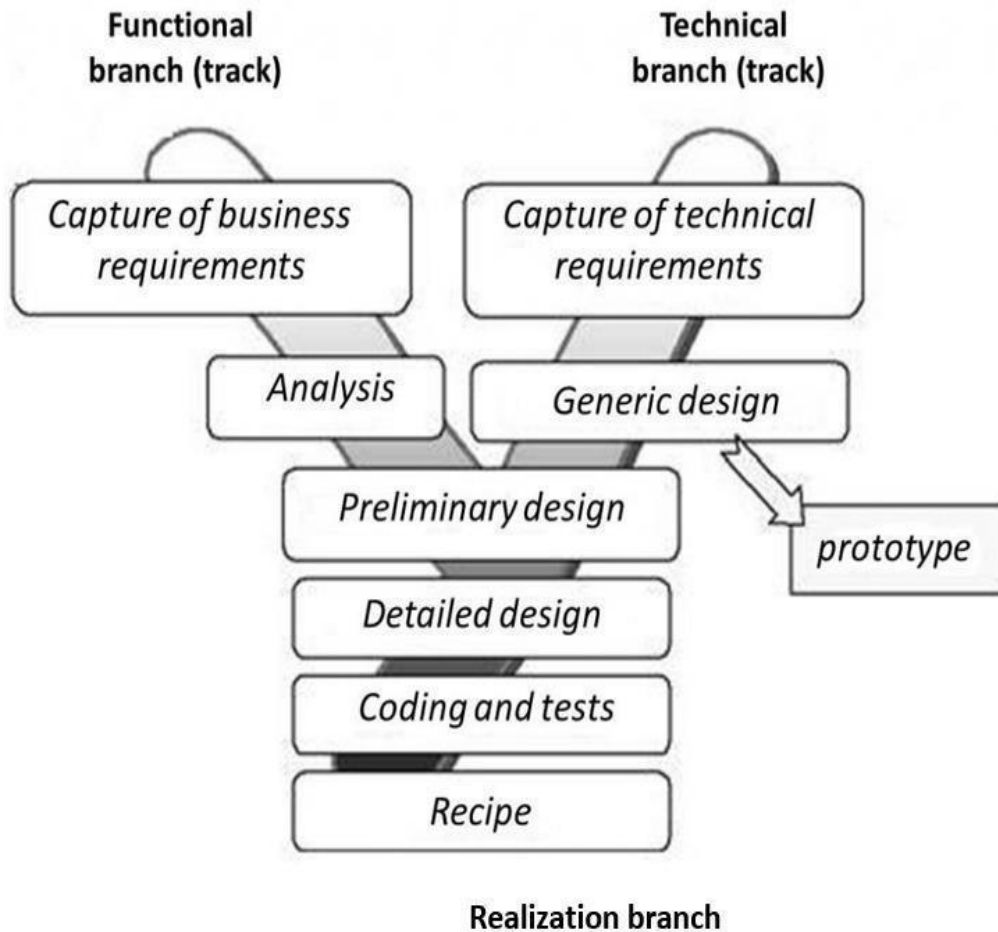


Figure 5 2TUP diagram

i.) The Functional Branch:

This branch captures business needs, resulting in a model focused on the end-user's requirements.

ii.) The Technical Branch:

This branch proposes a generic design from the technical needs, identifies the materials, tools, and technologies to be used.

iii.) The Implementation:

The implementation phase consists of bringing the two branches together, providing the application design, and finally delivering a solution adapted to the needs.

a) Functional Branch (left branch)

- ❖ Capture of Business Needs: This part minimizes the risk of producing an inadequate system with the needs of users and also verifies its consistency.
- ❖ Analysis: This is the study of the specifications to find out what the system will be made of in terms of the business rules. We have the following diagrams: Use Case, Activity Diagram, State Machine Diagram, and Communication Diagram.



b) Technical Branch

- ❖ **Capture of Technical Needs:** This stage consists of the identification of tools, materials, and technologies to use in developing the system. This technical architecture will be presented in this stage.
- ❖ **Generic Designs:** The technical architecture will be presented in this stage. We have the following diagram: mockUp7 Design.

c) Implementation Branch (middle branch):

- ❖ **Preliminary Branch:** This is the stage where the analysis model is integrated into the technical architecture. The goal here is to know what technical component will be used depending on the features of the analysis. We have the following diagrams: interaction overview, Component, Deployment, package diagram and Composite Diagram.
- ❖ **Detailed Design:** this is the detailed design of each feature of the system. We have the following diagrams: class, object, sequence and timing Diagrams.
- ❖ **Coding and Testing:** This is the programming phase of the designed features, alongside testing of the coded features.
- ❖ **Recipe (Result):** This is the validation phase of the functions of the system developed.

c. CHARACTERISTICS OF 2TUP

The two-track unified process was built to be used with the Unified Modeling Language to become a method and has the following characteristics:

- ❖ **It is user oriented:** 2TUP is built from user's expectation
- ❖ **It is component-oriented:** it offers flexibility to the model by supporting the re-use of components.
- ❖ **It is an iterative process:** processes are being done in iterations, and each iteration shows a precise level of abstraction.
- ❖ **It is an incremental process:** allowing a better functional and technical risk management and thus constituting the deadlines and the cost control.

d. JUSTIFICATION OF THE CHOICE OF THE ANALYSIS

Our choice was based on the modeling language UML, associated with it 2TUP, due to certain criteria, which are as follows:

- ❖ UML is based on an object-oriented approach;
- ❖ UML associates data and treats it;

- ❖ It is a modeling language that focuses mainly on users' needs and expectations;
- ❖ 2TUP is a process built from an object-oriented approach and is built from UML.
- ❖ 2TUP facilitates the modeling of complex systems, which can further be readjusted.

II. MODELLING

a. USECASE DIAGRAM

i. DEFINITION

The use case diagram simply shows the relationship between the actors of the system and the system. It permits users to easily visualize the different actions that can be done by the users. It also clearly defines the boundaries of the system.

ii. FORMALISM

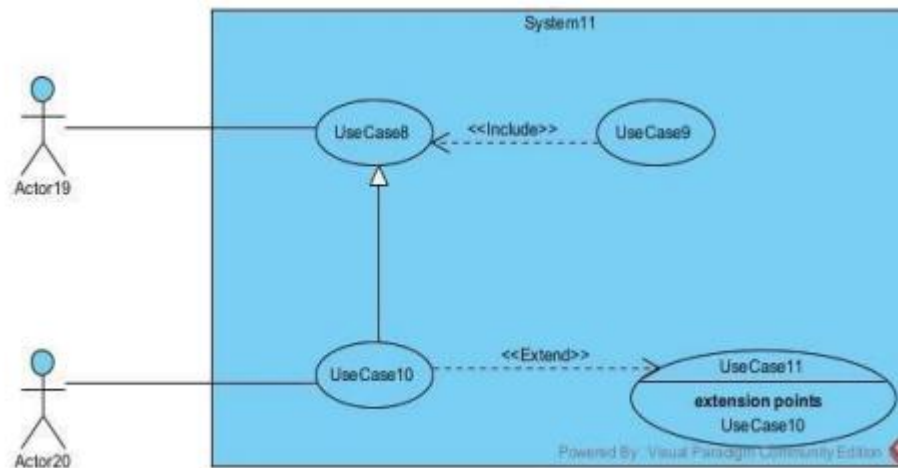
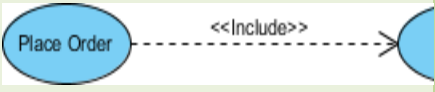
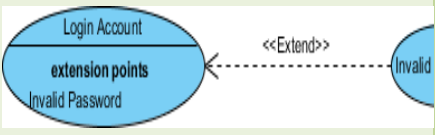


Figure 6: Formalism of use case diagram

iii. COMPONENTS OF USE CASE DIAGRAM

Table 11 Component of use case diagram

Element	Notation	Description
Actors		Represents an entity that directly interacts with the system. The actor is what performs the different possible actions of the system
Use case		A use case represents a functionality of the system. It is an action that can be performed by an actor.
Association		it indicates that an actor takes part in a use Case
Include		An inclusion denotes that an included action must be performed before the including action can be performed.
Extend		An extension denotes that an extending action may be performed while an extended action is being performed.

Generalization		This shows that an actor or a use case is a kind of another abstract or concrete actors which can be defined and later specialized using generalization relationship.
System		It is a container of use cases which interact with external actors

iv. ACTORS OF OUR SYSTEM

➤ Primary Actors

Table 12 Actors of our system

Actor	Role
Patient	View different information displayed on the welcome page and can create an account.
Heath Expert	Health experts deliver medical support by reviewing patient health status submissions, responding to consultation requests via chat or video calls, providing personalized health recommendations, and updating crisis management protocols. They manage their profiles to ensure accurate credentials and availability for patient interactions.

Administrator	Admins ensure platform efficiency by registering and approving health experts, managing patient accounts (e.g., activating or deactivating), publishing and validating health tips, monitoring consultation records, tracking payments processed through the Payment API, and generating platform usage reports.
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➤ Secondary actors

Table 13 Secondary Actors of Pharmaconnect

Actor	Role
Payement API	This external system processes financial transactions by validating payment information, executing secure payments for patient consultations, and sending confirmation receipts to the app, enabling seamless and secure consultation bookings.
Chatbot API	This external system provides an interactive, conversational assistant within the app. It helps families quickly access information about sickle cell disease, genotype compatibility, and preventive measures without needing to search through multiple sections.

V. GENERAL USE CASE DIAGRAM

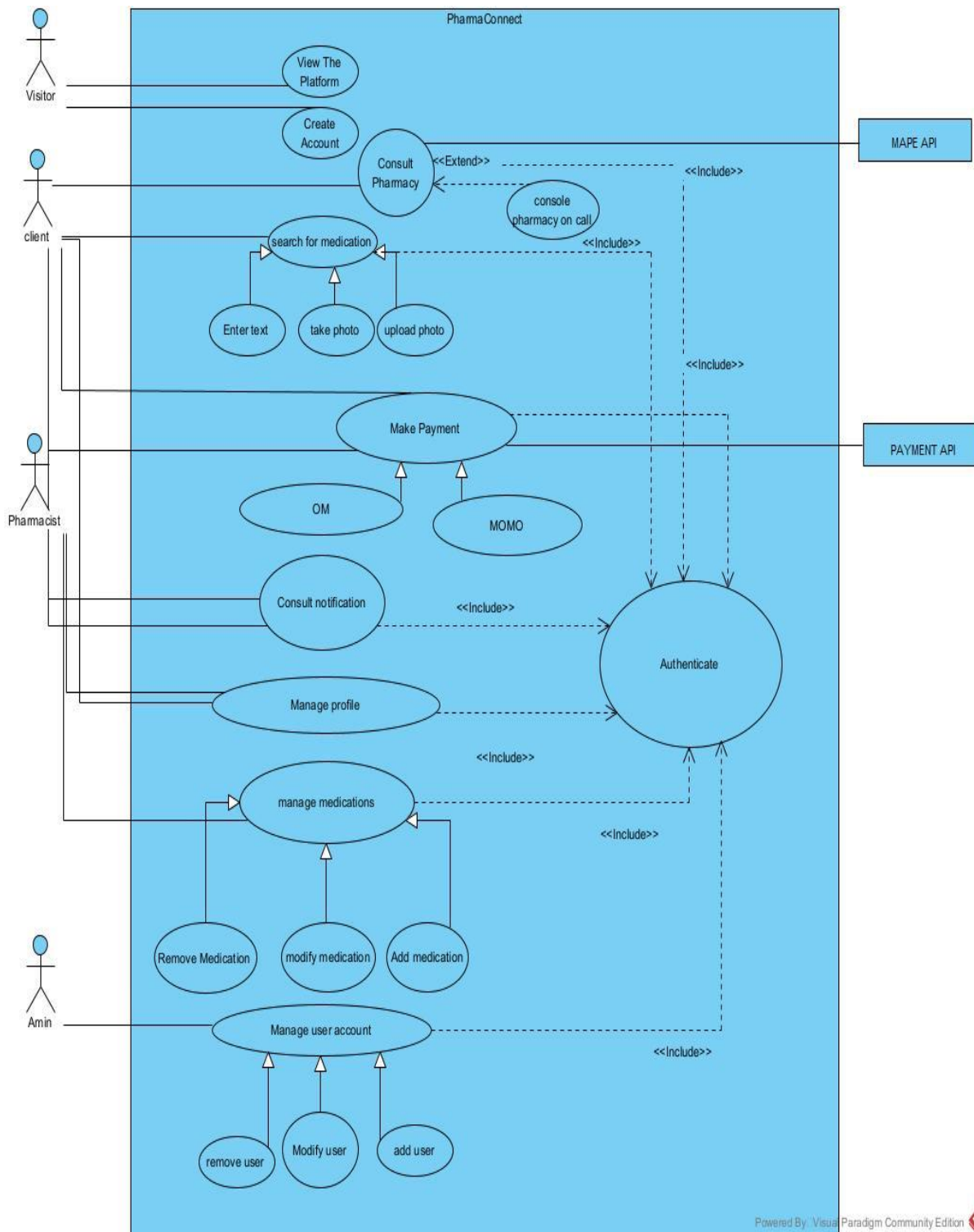


Figure 7 General Use case of pharmaconnect

vi. SPECIFIC USE CASE DIAGRAM FOR MAKE GENOTYPE MATCH TEST

Figure 8 Specific use case diagram for make payment

vii. TEXTUAL DESCRIPTION OF USE CASES

➤ Authenticate

Table 14 textual description of use case

USECASE AUTHENTICATE	
Summary	Permit Patient to access the system
Actors	Client / pharmacist
Pre-condition(s)	The actor already has an account; The app is launched The account must be active
Trigger	The user clicks on login button

Nominal scenario	1. The user clicks on the “Search Drug button in the mobile application. 2. The system displays a search form requesting the drug name. 3. The user enters the drug name and submits the request. 4. The system verifies the input for correctness (e.g., valid drug name). 5. The system sends a query to the database and checks available pharmacies. 6. The database verifies drug availability and pharmacy stock levels. 7. The database sends the response back to the system with pharmacy details. 8. The system displays a list of nearby pharmacies that have the drug, along with their locations.
Alternative scenario	The system displays an error message if the credentials entered by the Land owner does not match any land title existing in the database.
Post-condition of success	The user logs in successfully
Post condition of failure	An error message is display and user does not have access his space.

➤ Make payment

Table 15 Textual description of make payment

Actions	Patient
Summary	Permit patients to pay for requesting a consultation
Actors	Patient, Pharmacist, Payment API
Pre-condition(s)	The app is launched The customer has successfully searched or requested a drug The Pharmacist expert has confirmed the Request
Trigger	The user clicks on payment button
Nominal scenario	The user clicks on payment button The system displays the different payment options The customer clicks on preferred payment The system displays payment form The customer fills form and submit The system checks conformity The system sends payment request The system treats the result returned by the payment API The system sends "Update booking status to "Paid" to the database The system treats the result returned by the database The system displays a successful message
Alternative scenario	If step 6 of the nominal scenario the format is not correct, an error message is returned to the customer and he returns to step 4 of the nominal scenario.



- . If step 8 of the nominal scenario an error is found, an error message is returned to the customer and he returns to step 4 of the nominal scenario
- . If step 10 of the nominal scenario an error is found, an error message is returned to the customer and he returns to step 4 of the nominal scenario

Post-condition of success

If successful, the booking is marked as "Paid"

The logistic provider is notified that the booking has been paid for.

The system keeps a record of the payment transaction.

Post condition of failure

An error message is displayed

b. COMMUNICATION DIAGRAM

i. DEFINITION

The communication diagram is used to model the specific interaction between the participants of the system. It describes both the static and dynamic behavior of the system. It shows the chronological succession of the actions performed by an actor to realize an operation.

ii. FORMALISM

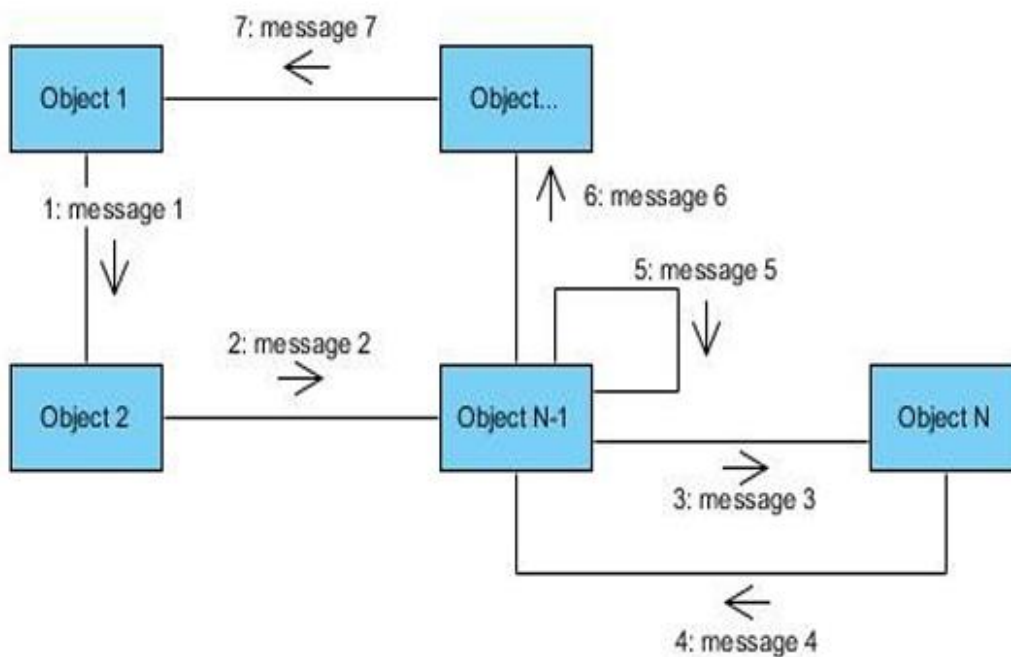


Figure 9 Formalism of communication diagram

iii. COMPONENTS OF COMMUNICATION DIAGRAM

Table 16 Components of communication diagram

Element	Notation	Description
Message		Designs a particular communication between lifelines.
Actors		A role plays by an entity that interacts with the subjects
Link		It initiates an association it connects two objects together for them to communicate.
Object		An actor represents an individual participant in the interaction conversation

iv. AUTHENTICATE COMMUNICATION DIAGRAM

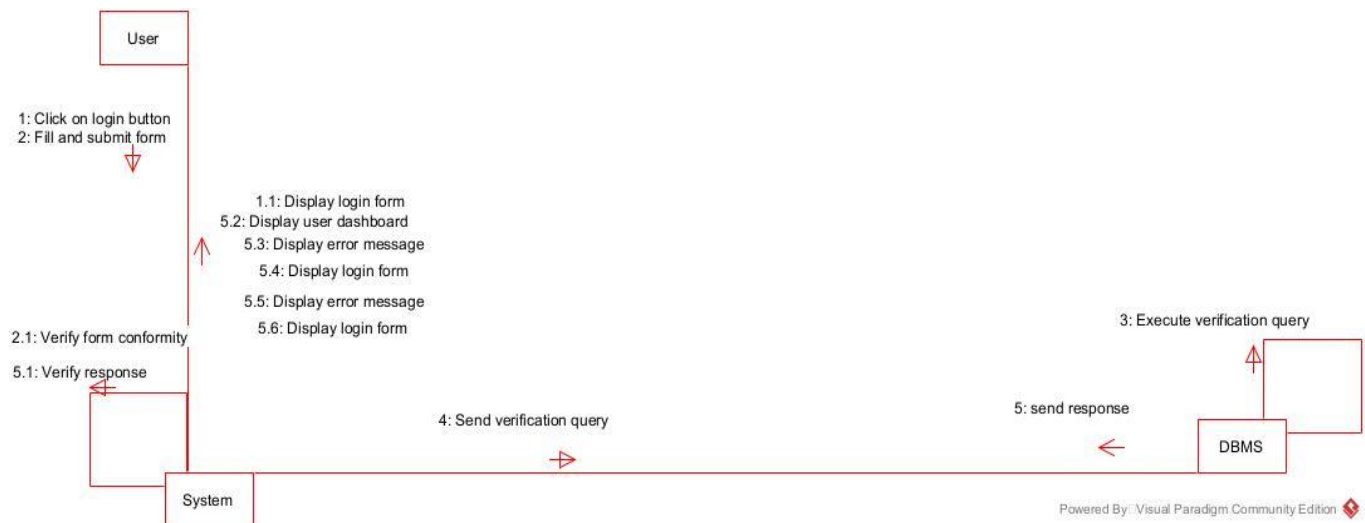


Figure 10 Authenticate communication diagram

v. COMMUNICATION FOR DIAGRAM FIR DRUG REQUEST

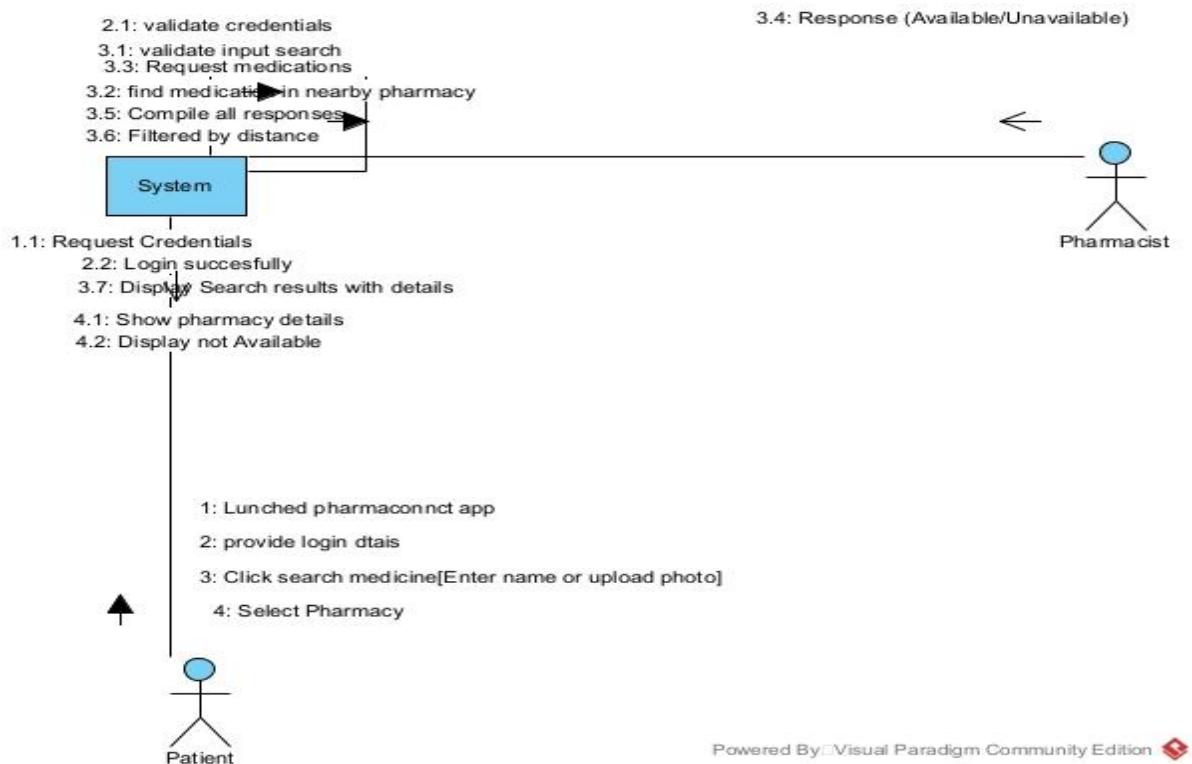


Figure 11 Medication request match communication diagram

vi. MAKE PAYMENT COMMUNICATION DIAGRAM

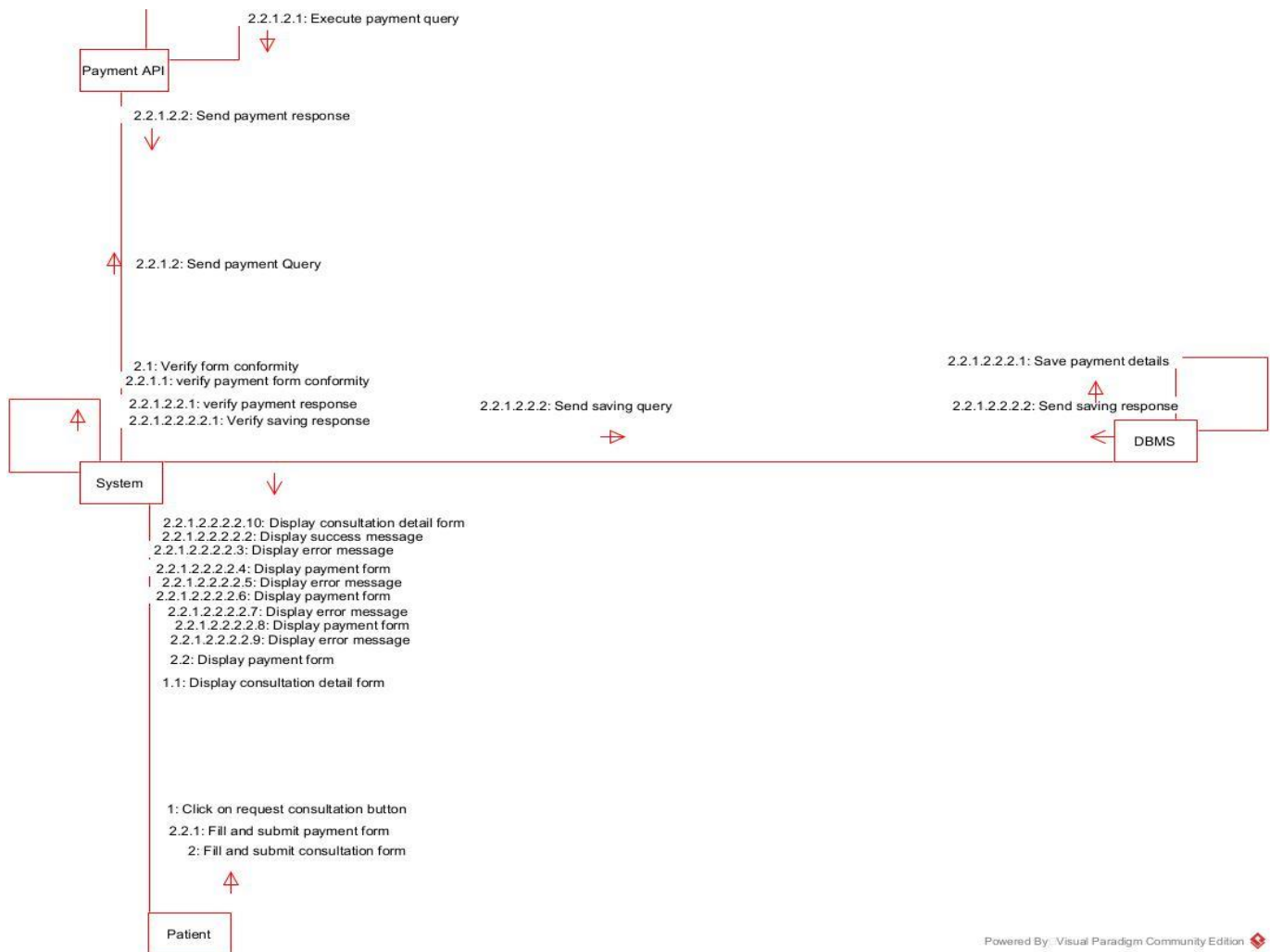


Figure 12 make payment communication diagram

c. SEQUENCE DIAGRAM

i. DEFINITION

A sequence diagram is a UML diagram that illustrates the sequence of messages between objects in an interaction. It consists of a group of objects that are presented by lifelines and the messages that they exchange over time during the interaction. These interactions are arranged from top to bottom following their order of occurrence.

ii. FORMALISM

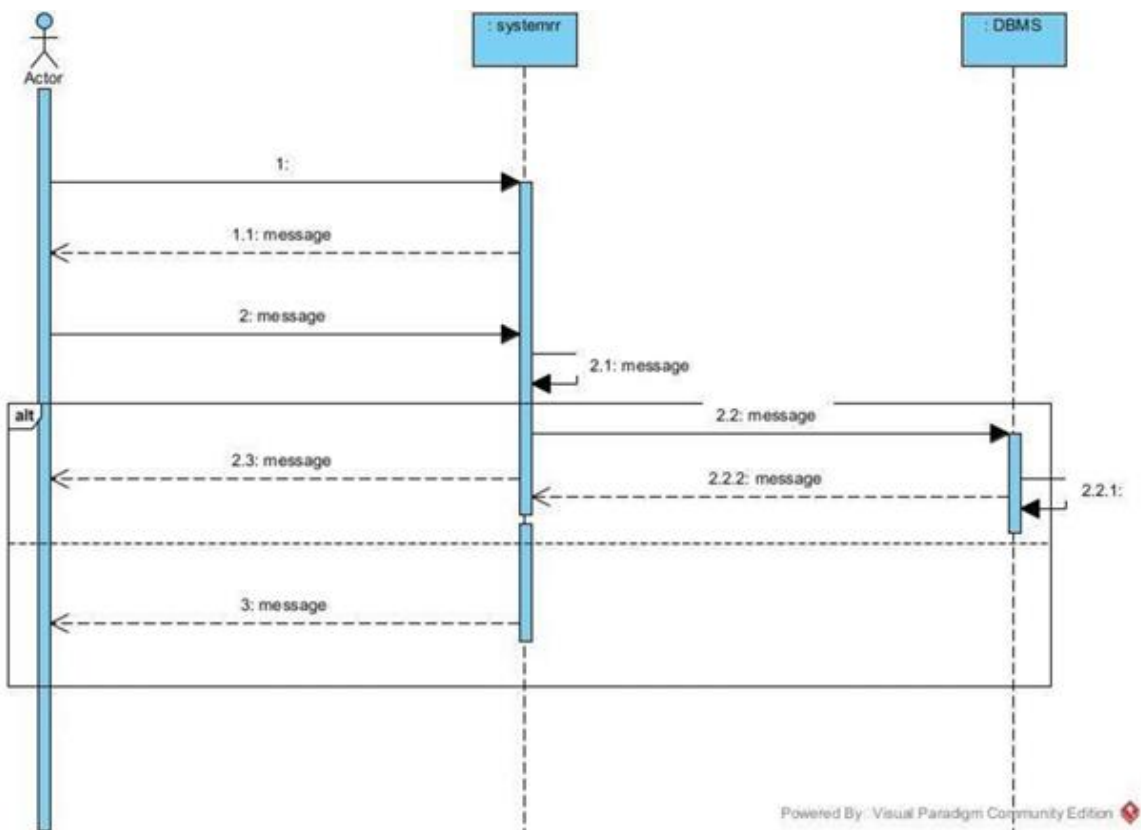
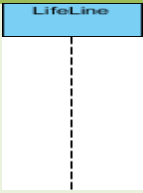
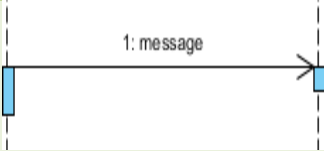
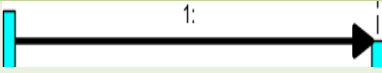
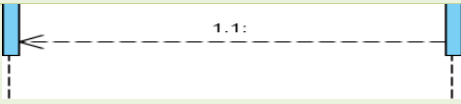
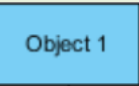


Figure 13 Formalism of sequence diagram

iii. COMPONENTS OF SEQUENCE DIAGRAM

Table 17 Components of sequence diagram

Element	Notation	Description
Lifelines		They represent rows or objects instances that participate in the sequence being modelled.
Asynchronous Message		It is a message that receives an indirect response.
Synchronous Message		It is a message that sends and want response before it continues a process
Self-Message		Self-message is a kind of message that represents the invocation of message of the same lifeline.
Return Message		It represents the response of a message.
Actor		They send and receive message.
Combined Interaction on Fragment		An articulation of interaction diagram, defined by an operator and operands.
Object		They send and receive messages

iv. CREATE ACCOUNT SEQUENCE DIAGRAM

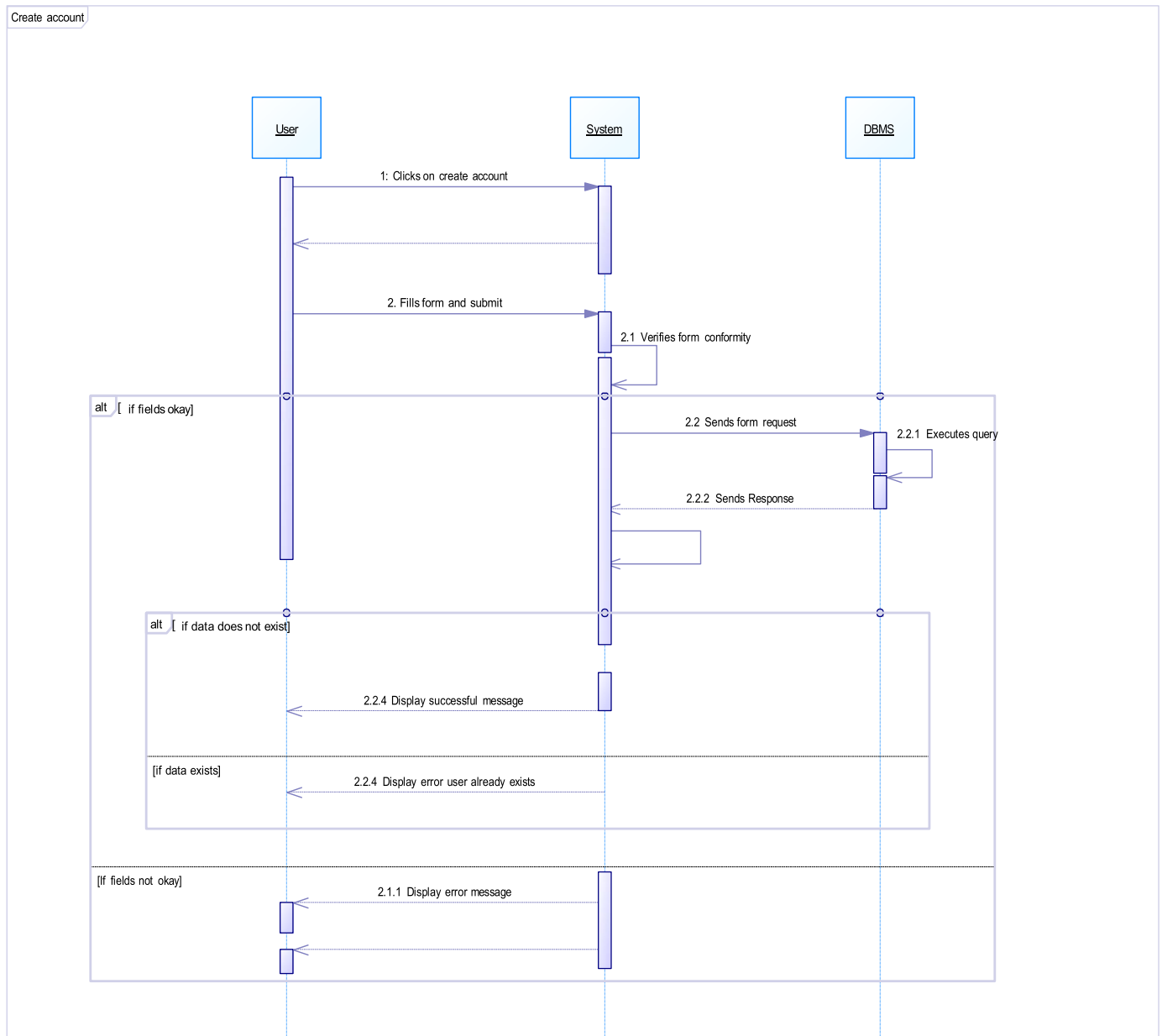


Figure 14 Authentication sequence diagram

Authentication Sequence and diagram

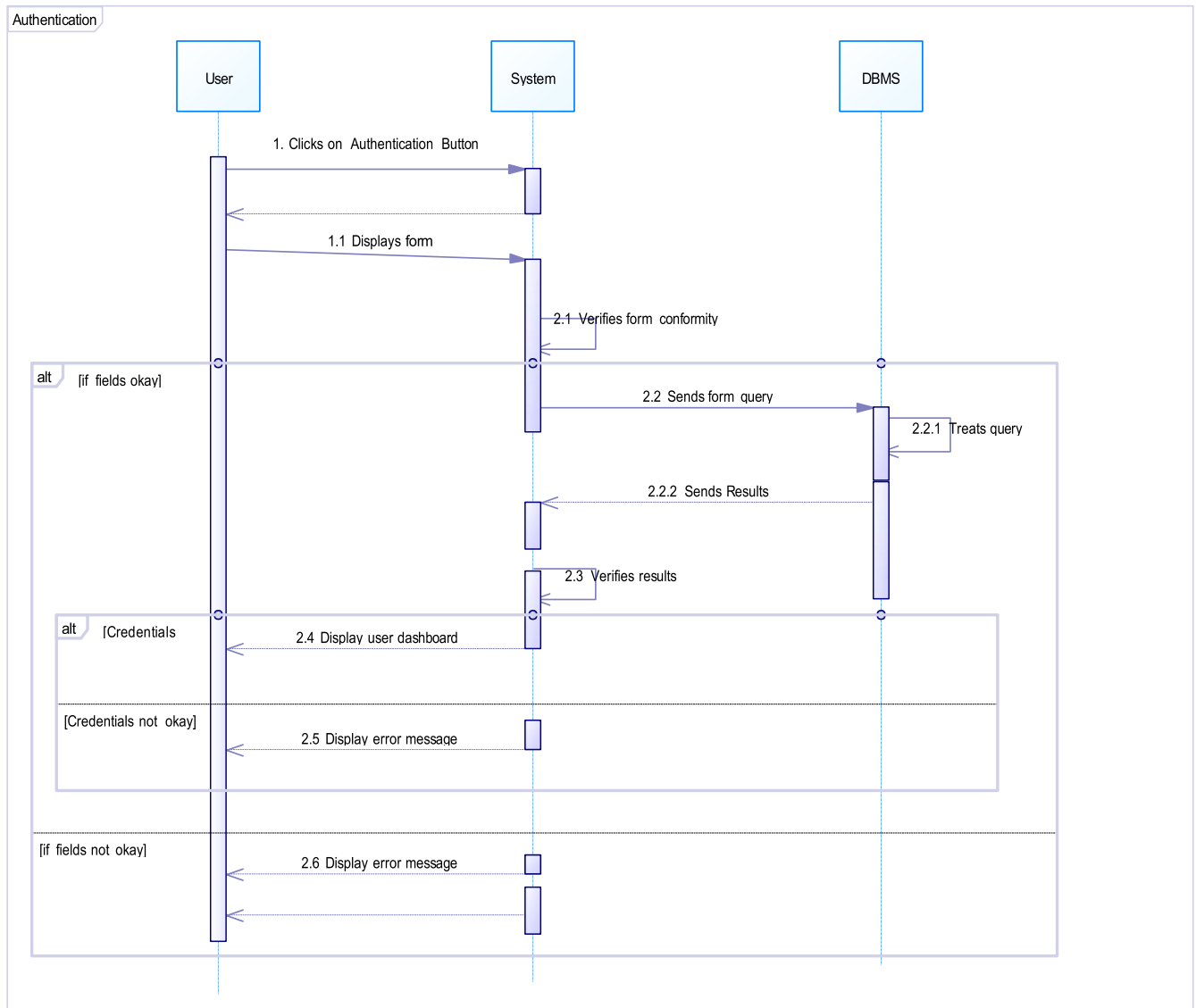


Figure 15 make Authentication sequence diagram

v. Sequence diagram for medication request

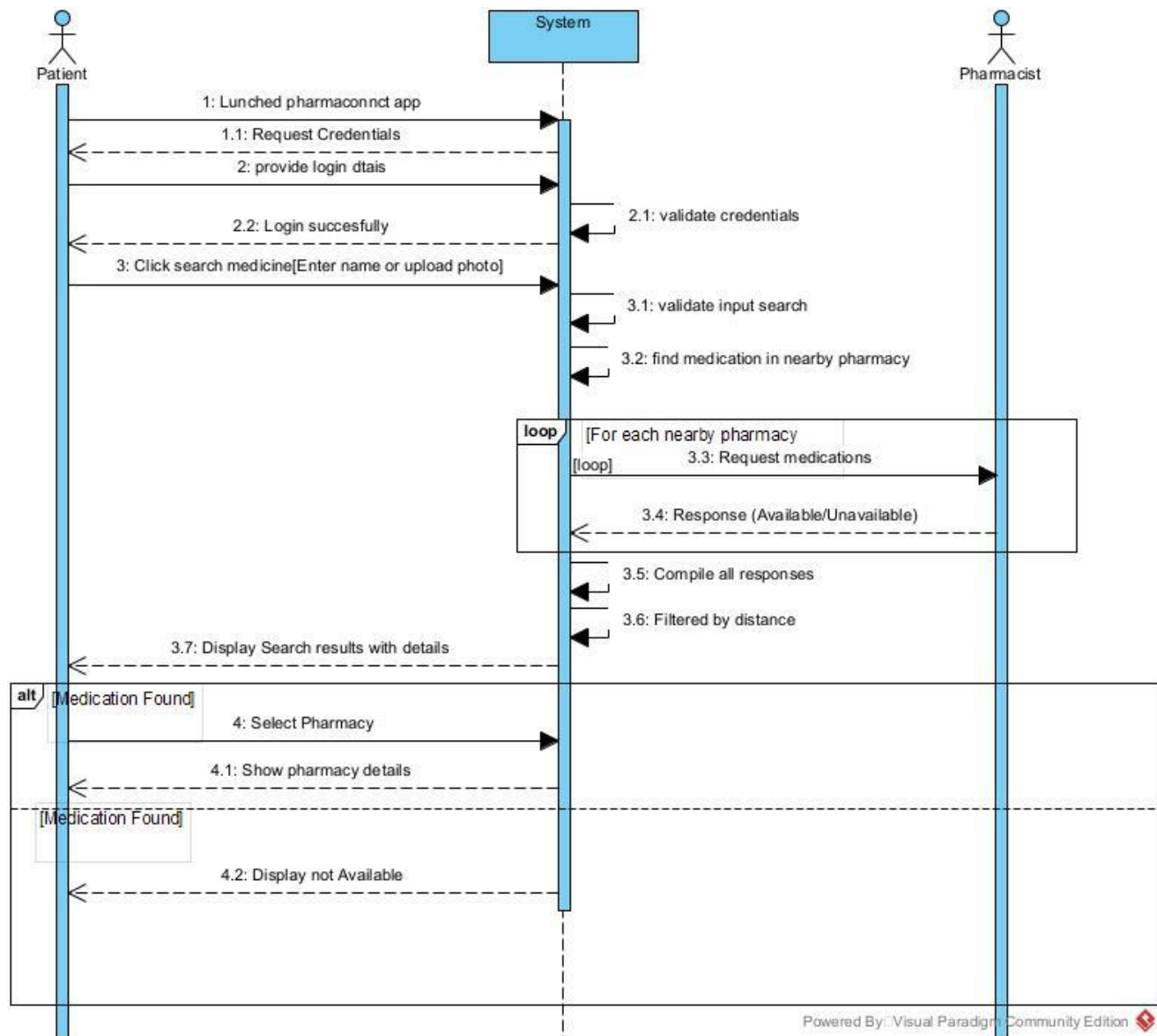


Figure 16 sequence diagram for medication request

d. ACTIVITY DIAGRAM

i. DEFINITION

An activity diagram is a graphical representation of workflows that show the steps needed in the realization of a process, showing the details from a start point to an end point through all decisions and actions that can possibly be performed. Activity diagrams are intended

to model both the computational and organizational processes. The flow can be sequential, branched, or concurrent. Below is an activity diagram formalism.

ii. FORMALISM

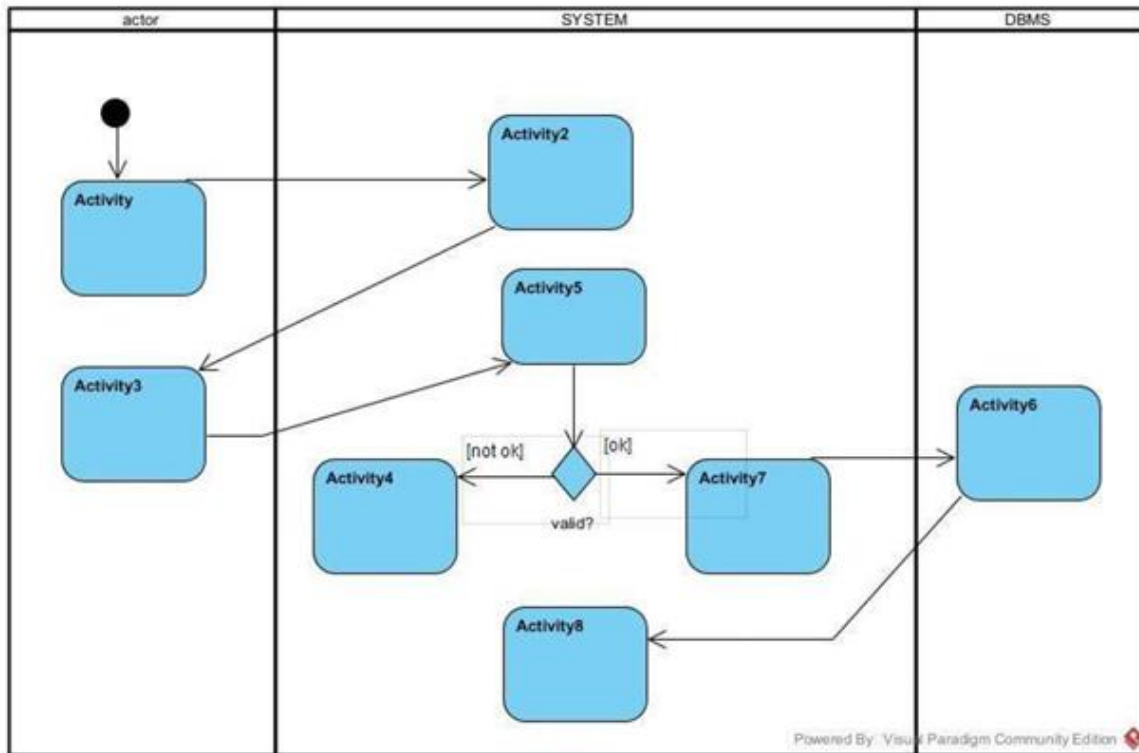
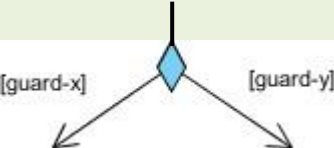


Figure 17 Formalism of activity diagram

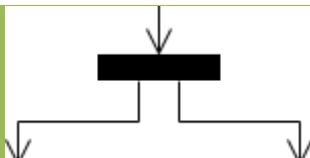
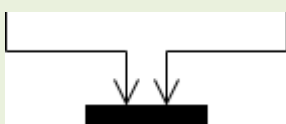
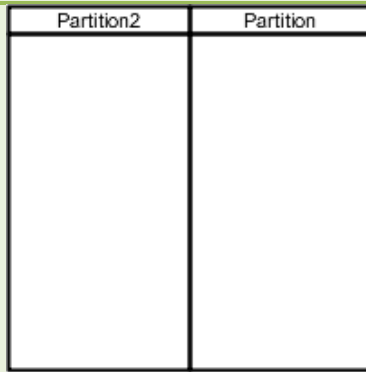
iii. COMPONENTS OF ACTIVITY DIAGRAM

Table 18 Components of activity diagram

Element	Notation	Description
Activity		Is used to represent a set of actions

Action		A task to be performed
Activity Edge		An activity edge in a UML activity diagram is a directed connection between two activity nodes. It represents the flow of control or data between these nodes.
Initial node		Portrays the beginning of a set of actions or activities
Activity Final Node		Stop all control flows and object flows in an activity (or action)
Object node		Represent an object that is connected to a set of Object Flows
Decision node		Represent a test condition to ensure that the control flow or object flow only goes down one path

Merge node		Bring back together different decision paths that were created using a decision-
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Fork node		node. Split behavior into a set of parallel or concurrent flows of activities (or actions)
Join node		Bring back together a set of parallel or concurrent flows of activities (or actions).
Swimlane and Partition		A way to group activities performed by the same actor on an activity diagram or to group activities in a single thread

iv. ACTIVITY DIAGRAM FOR CREATE ACCOUNT

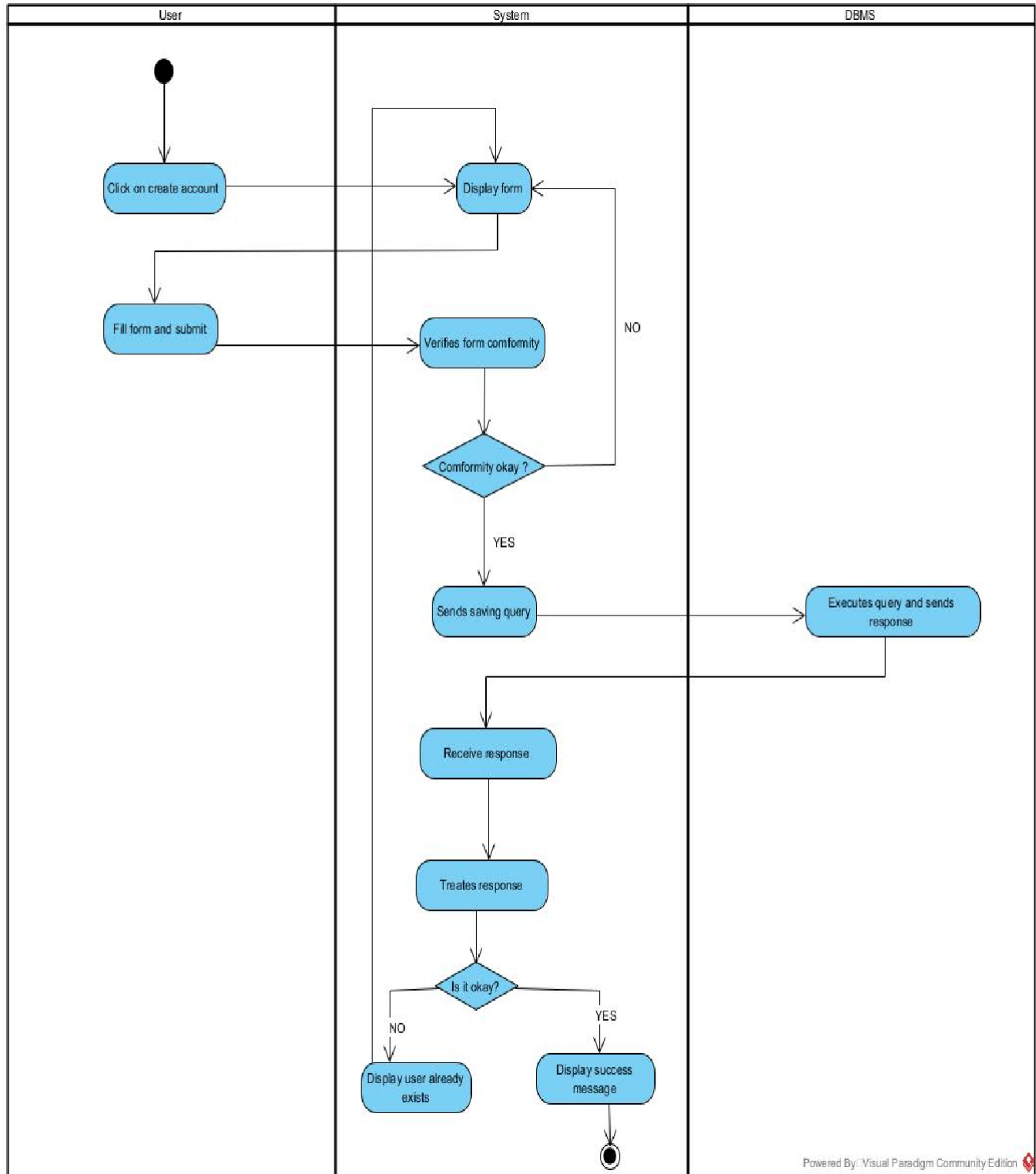


Figure 18 Authenticate activity diagram

V. ACTIVITY DIAGRAM FOR AUTHENTICATION

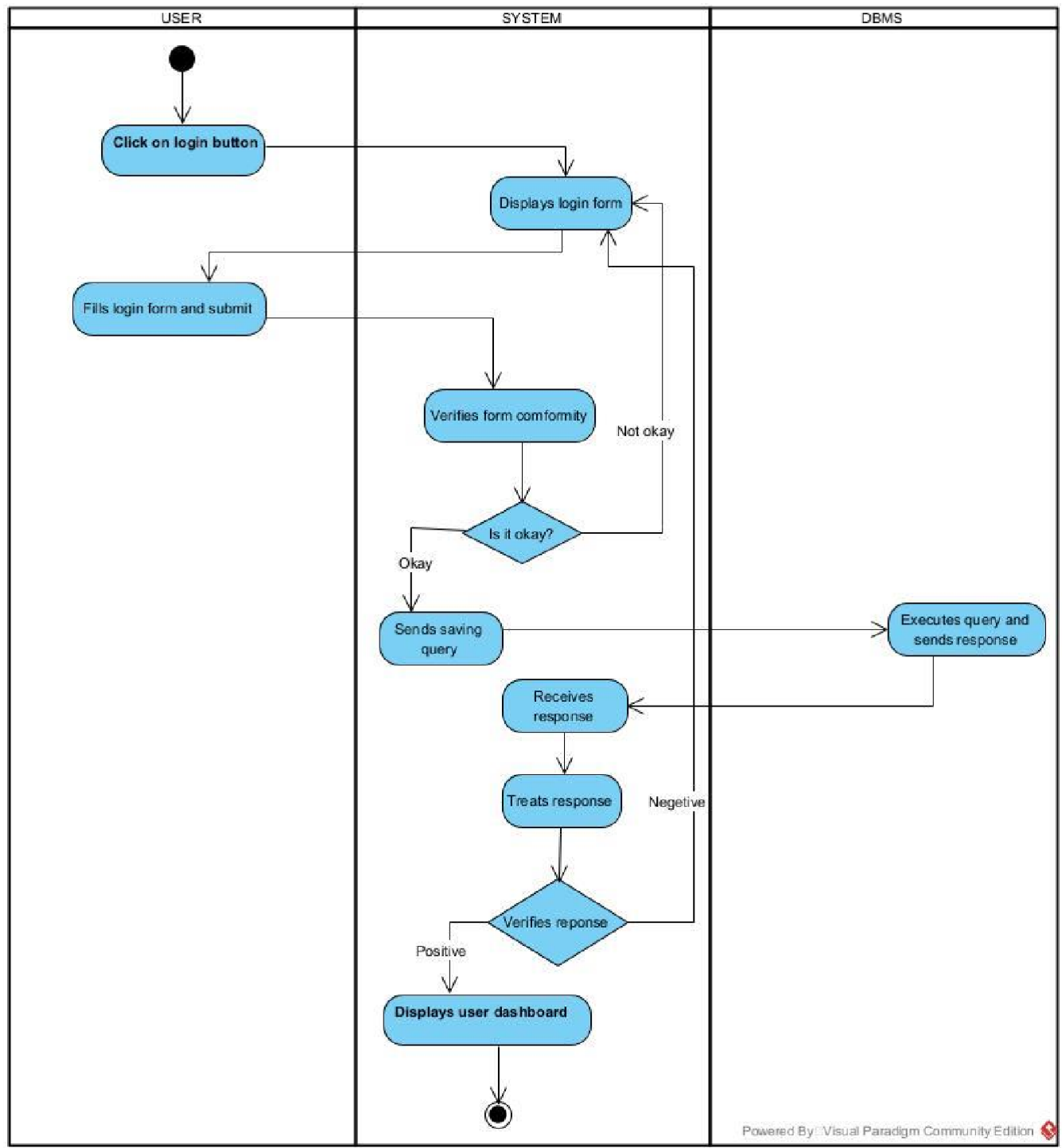


Figure 19 activity diagram for authentication

vi. ACTIVITY DIAGRAM FOR DRUG REQUEST

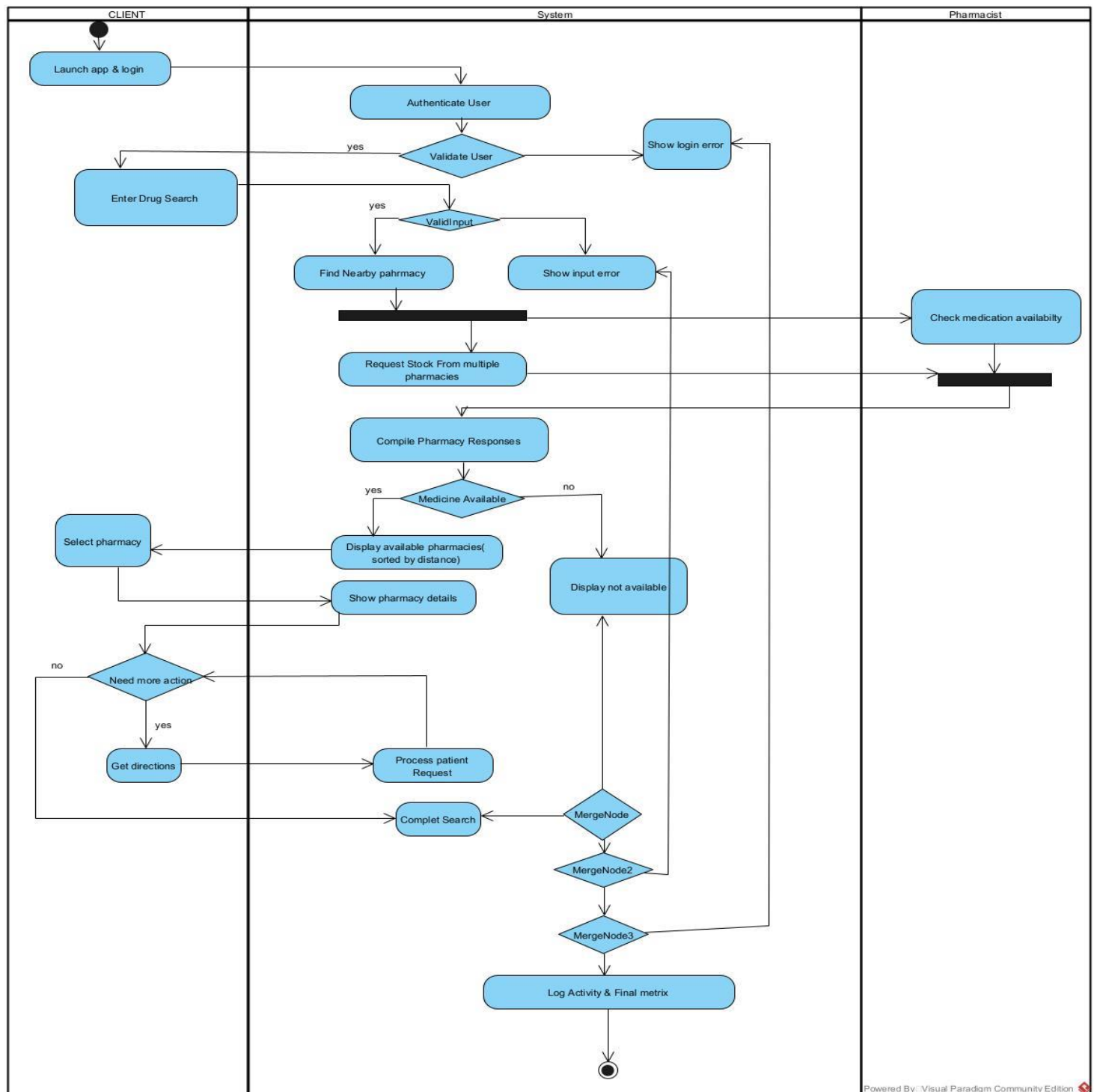


Figure 20 activity diagram for drug request

CONCLUSION

Having reached the end of our analysis, we were asked to illustrate the language and method used during this phase, present the different diagrams of the functional branch. we can then say that this analysis file has allowed us delimit the functional needs of the web application that will be designed and to have a detailed overview of the new system to be set up without notwithstanding we quickly dive into the conception phase.



FILE 4 : CONCEPTION PHASE

Preamble

This document makes it possible to model the proposed solution as a whole and to gather the information necessary to set up a complex and efficient database. It therefore foresees the future system.

Content

INTRODUCTION

- I. CAPTURE OF TECHNICAL NEEDS
- II. CLASS DIAGRAM
- III. STATE MACHINE DIAGRAM
- IV. PACKAGE DIAGRAM

CONCLUSION

INTRODUCTION

The realization of an analysis file is very important for the advancement of the application; it guarantees the reliability of maintenance and use of a software. That said, after analyzing our system, we will move on to the design of our future system. To do this we will follow the steps of the technical branch of the 2TUP process, making the diagrams associated with these steps namely the class diagram and the state machine diagram.

I. CAPTURE OF TECHNICAL NEEDS

i. PHYSICAL ARCHITECTURE

We implemented a three-tier architecture system.

The three-tier architecture is a client-server software architecture that consists of the user interface (presentation tier), business rules (logic tier), and data access/storage tier (data tier).

- ❖ **Presentation tier:** It is the user interface and communication layer of the application, where the end user interacts with the application. Its main purpose is to display information to and collect information from the user.
- ❖ **Logic tier:** The logic tier, also known as the application tier or middle tier, is the heart of the application. In this tier, information collected in the presentation tier is processed sometimes against other information in the data tier using business logic, a specific set of business rules. The logic tier can also add, delete or modify data in the data tier.
- ❖ **Data tier:** The data tier, sometimes called database tier, data access tier or backend, is where the information processed by the application is stored and managed. This can be a relational database management system such as PostgreSQL, MySQL, MariaDB, Oracle, DB2, Informix or Microsoft SQL Server, or in a NoSQL Database server such as Cassandra, CouchDB or MongoDB. In a three-tier application, all communication goes through the logic tier. The presentation tier and the data tier cannot communicate directly with one another.

ii. LOGICAL ARCHITECTURE

The logical architecture of an application refers to the structural design that gives as much detail as possible without defining the system technology or environment. The architecture we have chosen for our application is the Model-View- Controller (MVC). The MVC architecture divides an interactive application into 3 components which are:

The model: this component corresponds to all the data-related logic of the application. It directly manages the data, logic and rules of the application.

The view: It is used for all the UI logic of the application. It obtains its data from the model via the controller and each view has an association controller component.

The controller: it acts as an interface between Model and View components to process all the business logic and incoming requests, manipulate data using the Model component and interact with views to render final output. The MVC can be represented as follows:

MVC Architecture Pattern

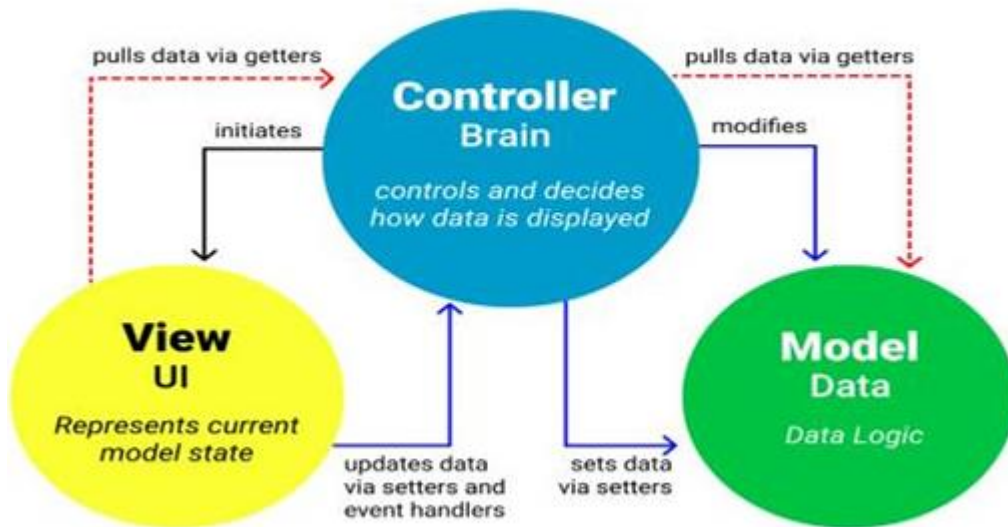


Figure 21 MVC architecture pattern

II. CLASS DIAGRAM

i. DEFINITION

A class diagram is a static diagram. It represents the static view of an application. class diagram is not only used for visualizing, describing, and documenting different aspects of the system, but also for constructing executable code of the software application. A class diagram describes the attributes and operations of a class and constraints imposed on the system. Its purpose is to model the static view of an application.

ii. FORMALISM

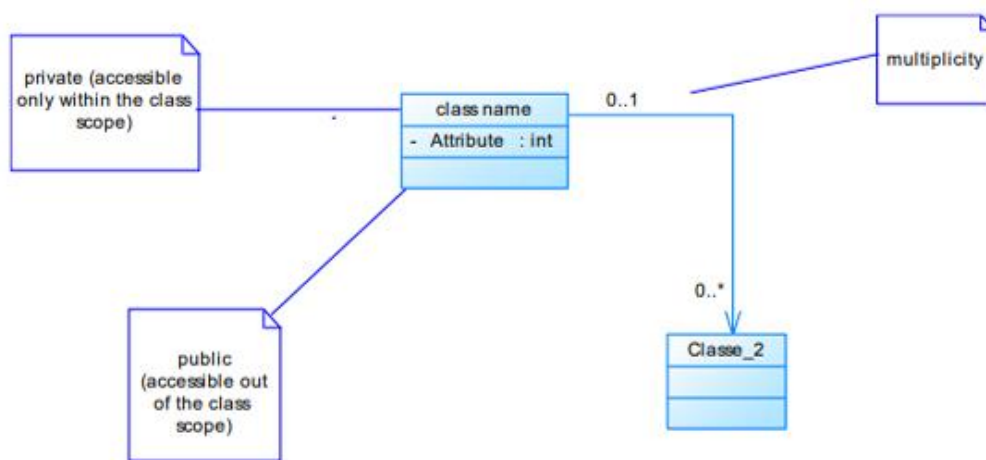
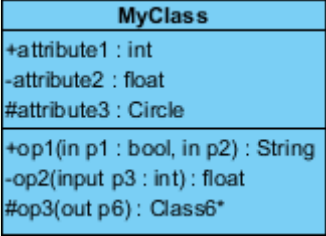
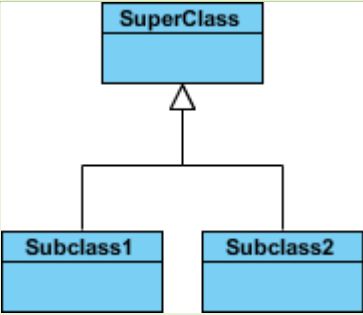
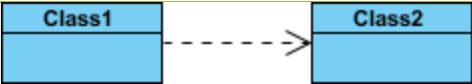


Figure 22 Formalism of a class diagram

iii. COMPONENTS OF CLASS DIAGRAM

Table 19 Components of a class diagram

Element	Notation	Description
Class		A class is an element that defines the attributes and behaviours that an object can generate
Generalization		it a relationship between a whole thing (called superclass) and a more specific thing (called subclass)
Association		It is a general type of relationship between elements, it may include cardinality, roles etc.
Aggregation		If the parent of the aggregate is deleted, the children are not deleted.
Composition		If a parent of a composite is deleted all its parts are deleted with it.
Dependency		It existed between two classes, if one changes it may cause the change in the order, but the other way around

iv. CLASS DIAGRAM OF DREPALIFE

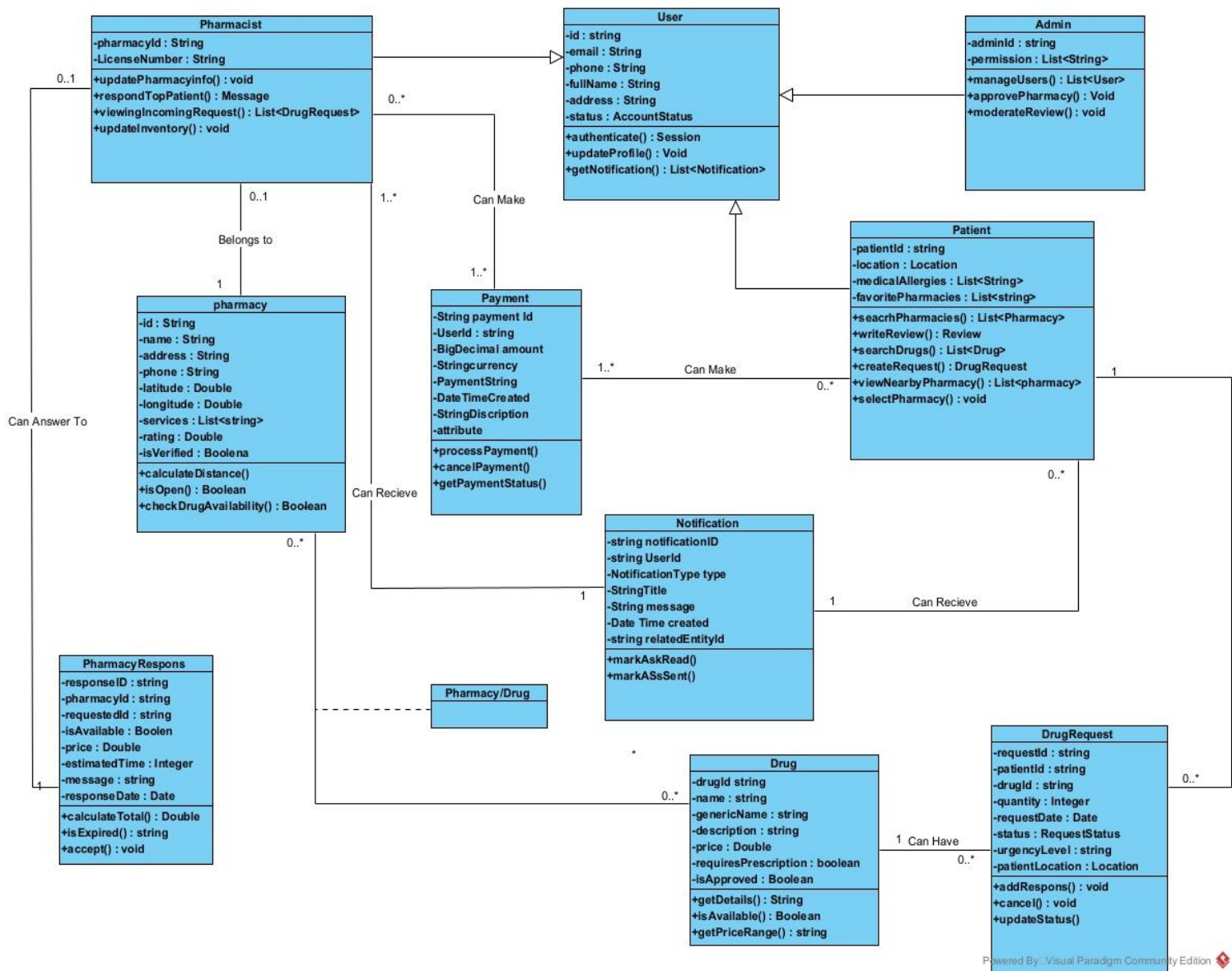


Figure 23 Class Diagram

III. STATE MACHINE DIAGRAM

i. DEFINITION

A state machine diagram describes the behavior of a single object in response to a series of events in a system. Also known as the state machine diagram, it models the dynamic flow of control from the state of a particular object within a system.

ii. FORMALISM

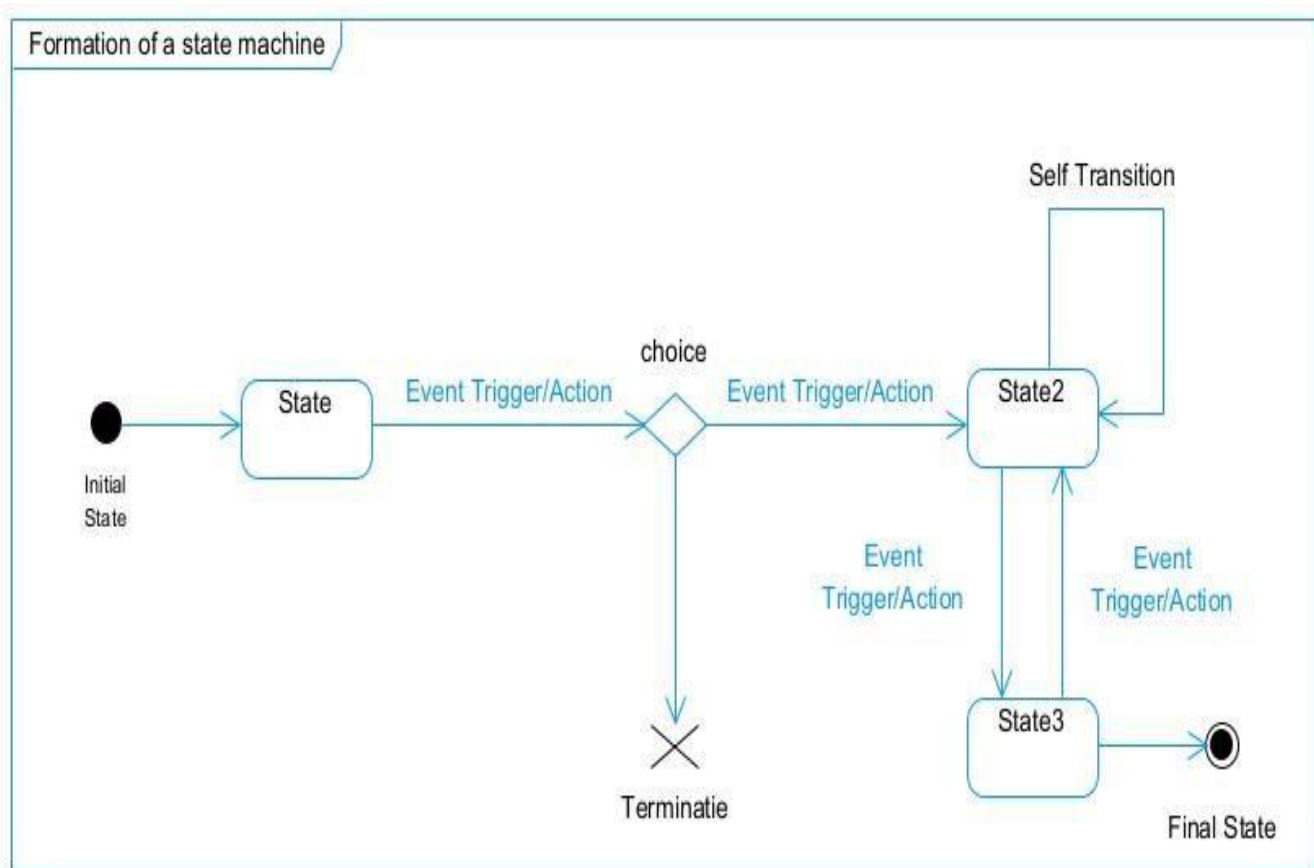
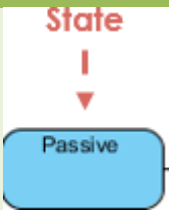


Figure 24 Formalism of state machine diagram

iii. COMPONENTS OF STATE MACHINE DIAGRAM

Table 20 Components of a state machine diagram

Element	Notation	Description
State		Models a situation during which a certain invariant condition holds.
First (Initial State)		It represents a default vertex, that is, a source for a single transaction to the default or composite state.
Final state		A state specifying that the enclosing region is complete
Transition		A direction relation between a source and a target vertex.
Choice pseudo state		A diamond symbol that indicates a dynamic condition with branched potential results
Terminate		Implies that the execution of a state by means of its context is terminated.

iv. ACCOUNT STATE MACHINE DIAGRAM

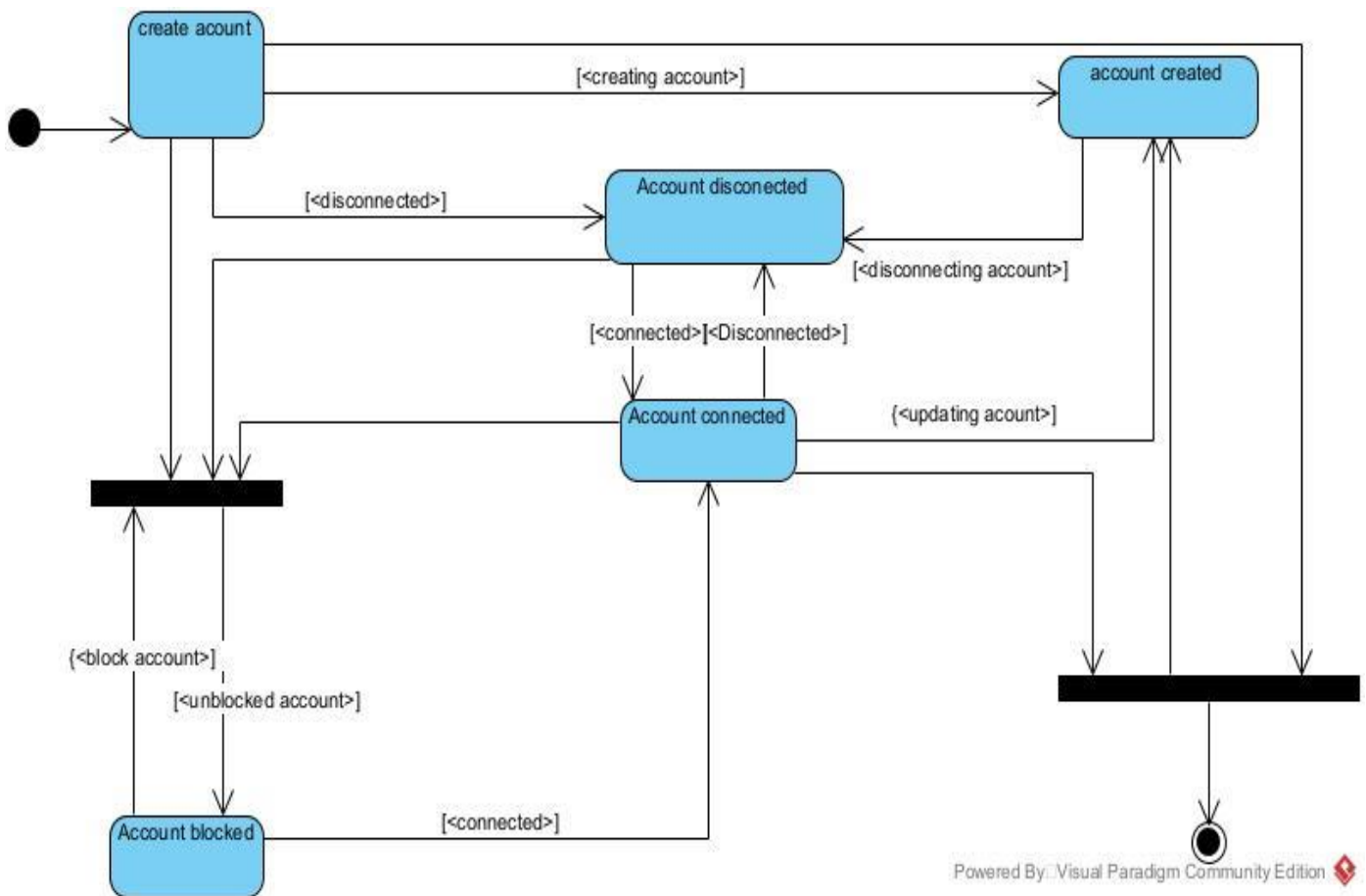


Figure 25 Account state machine diagram

IV. PACKAGE DIAGRAM

i. DEFINITION

A package diagram is a type of structural diagram in UML that represents the organization and dependencies of different packages within a system or software application. Package diagrams are useful for visualizing the overall structure and organization of a

system, as well as understanding the dependencies and relationships between the different components.

ii. FORMALISM

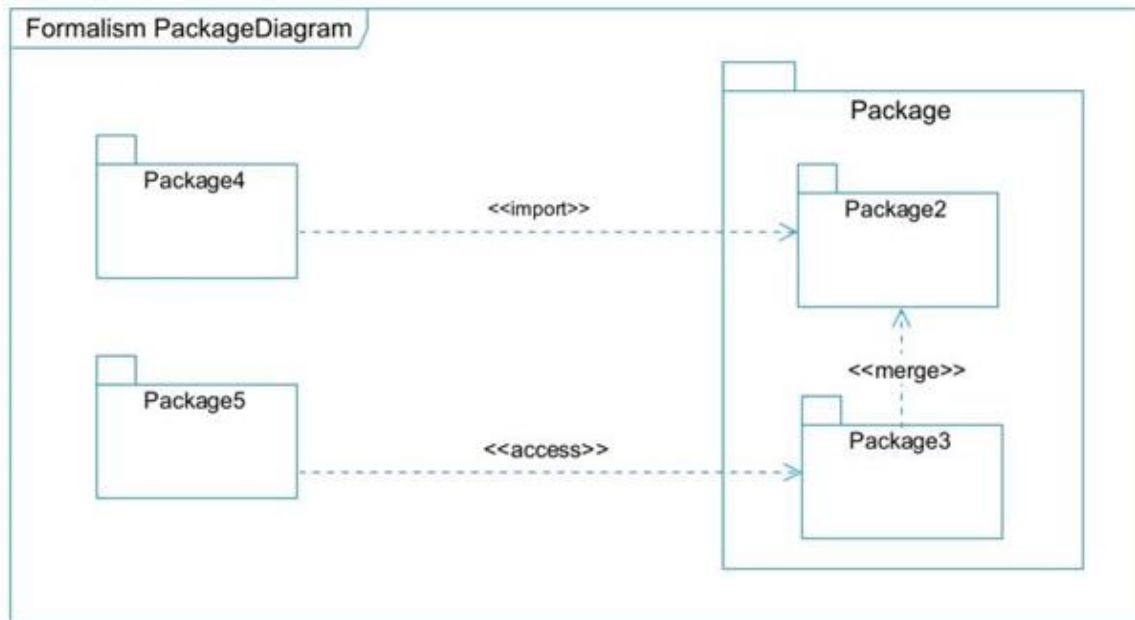
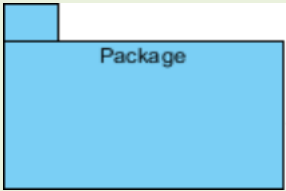
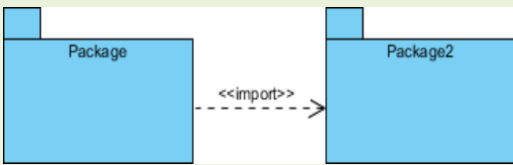
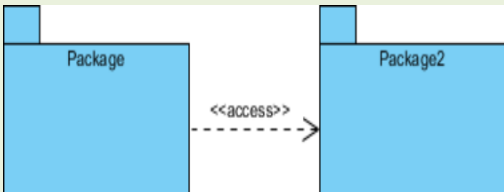


Figure 26 Formalism of package diagram

iii. COMPONENTS OF PACKAGE DIAGRAM

Table 21 Components of a package diagram

Element	Notation	Description
Package		A package is a namespace use to group related elements; it is a mechanism used to group elements into a better structure in a system.
Package import		A relationship Indicate that, functionality has been imported from one package to another.
Package access		A relationship Indicates that one package requires assistance from the function of another package.
Package merge		It is a relationship which shows that, the functionality of two packages are combines to a single function.

iv. PACKAGE DIAGRAM OF DREPALIFE

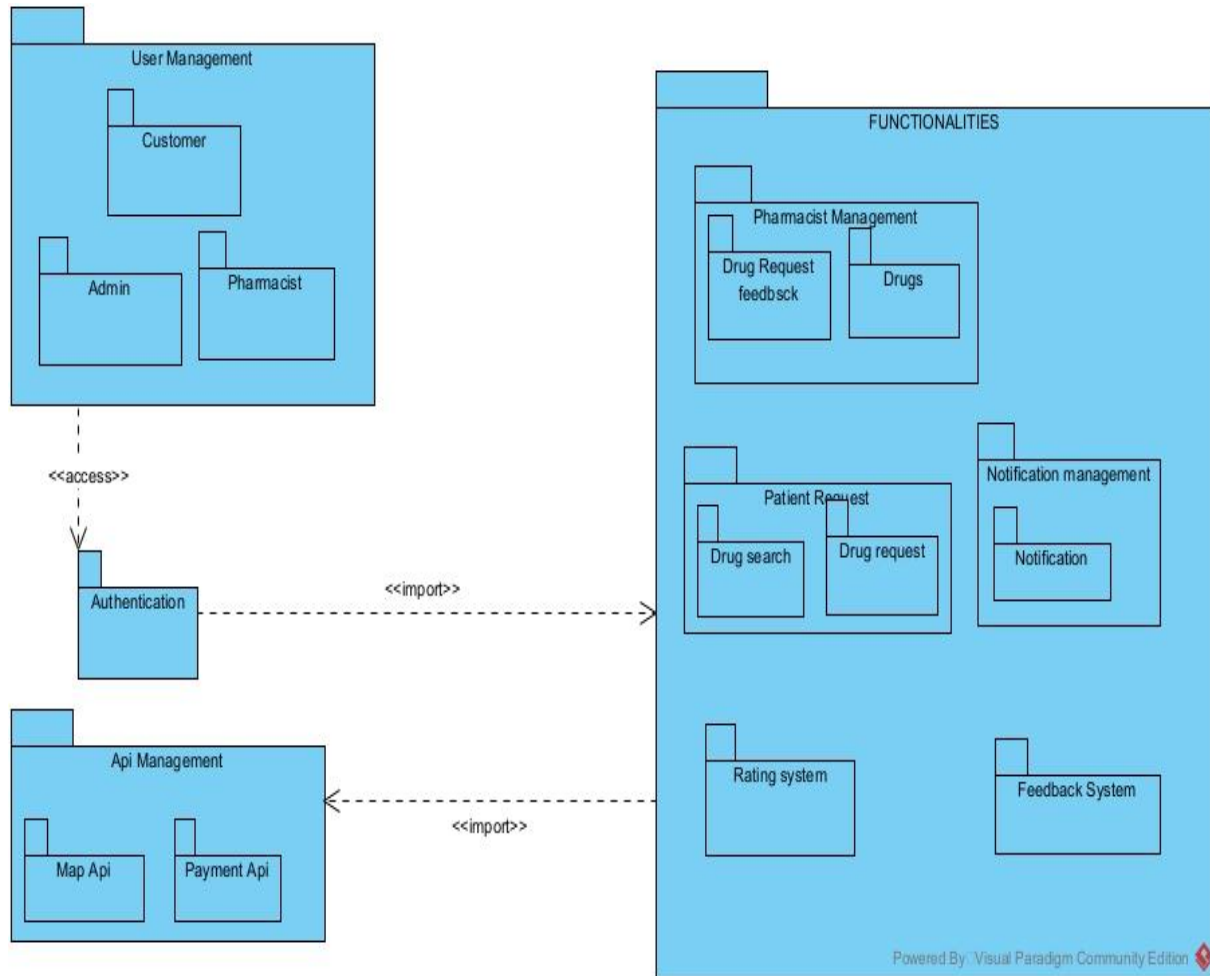


Figure 27 Pharma connects package diagram

CONCLUSION

In the conception phase, we set as objective to plans the different aspects of our system by showing how it will be structured and deployed within existing technical architectures. We began by considering the technical constraints for our system, after which we proceeded to identify the components of our system, how they are grouped together and how they should be deployed on appropriate deployment targets. We finished this phase by looking at interactions between the various aspects and actors of our system. The next phase of our report is the realization phase where we will look at aspects concerning the implementation of our system.



FILE 5 : REALIZATION

Preamble

The realization document consists of the implementation of the project in a programming language according to the specifications defined in the previous file. It contains component and deployment diagrams, the testing and development phases. At the end of this part, programming documentation will be produced explaining the architecture of the database and the architecture of our code.

Content

INTRODUCTION

- I. DEPLOYEMENT DIAGRAM
- II. COMPONENT DIAGRAM
- III. TECHNICAL CHOICES

CONCLUSION

INTRODUCTION

Design and modeling are important links in the development process of an application because they make it possible to define and specify the different elements constituting the application or system studied. Realization is the culmination of the work carried out upstream because it makes it possible to physically create the various objects that were designed, and to bring the computer system to life. Throughout this phase, we will present the software, physical, and hardware architecture of our system through diagrams such as the component diagram, deployment diagram, physical model, and code architecture.

I. DEPLOYMENT DIAGRAM

i. DEFINITION

The deployment diagram models the hardware components used to implement a system and the association between those components. Deployment diagrams can be implemented as early as the design phase to document the system's physical architecture. As elements used in deployment diagrams, we can have, components, as in component diagrams, and nodes, which represent the physical processing resources of the system, and their associations.

ii. FORMALISM

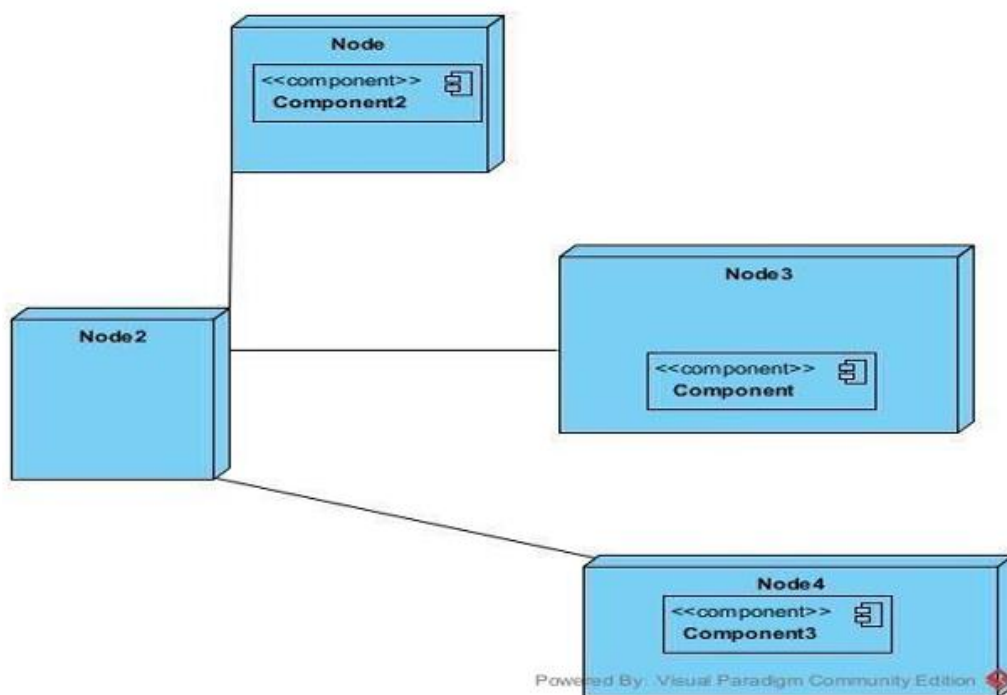
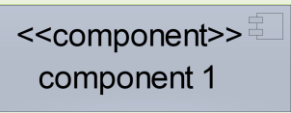


Figure 28 Formalism Deployment diagram

iii. COMPONENTS OF DEPLOYMENT DIAGRAM

Table 22 Component of the deployment diagram

Element	Notation	Description
Node		A hardware or software object, shown by a three-dimensional box
Artifact		An artifact is a major product, which is produced or used during the development of a software. E.g. diagrams, data models, setup scripts
Component		It represents a modular part of a system that encapsulates its content and whose manifestation is replaceable within its environment.
Association	association	An association helps to connect two nodes together which permits them to communicate together

iv. DEPLOYMENT DIAGRAM

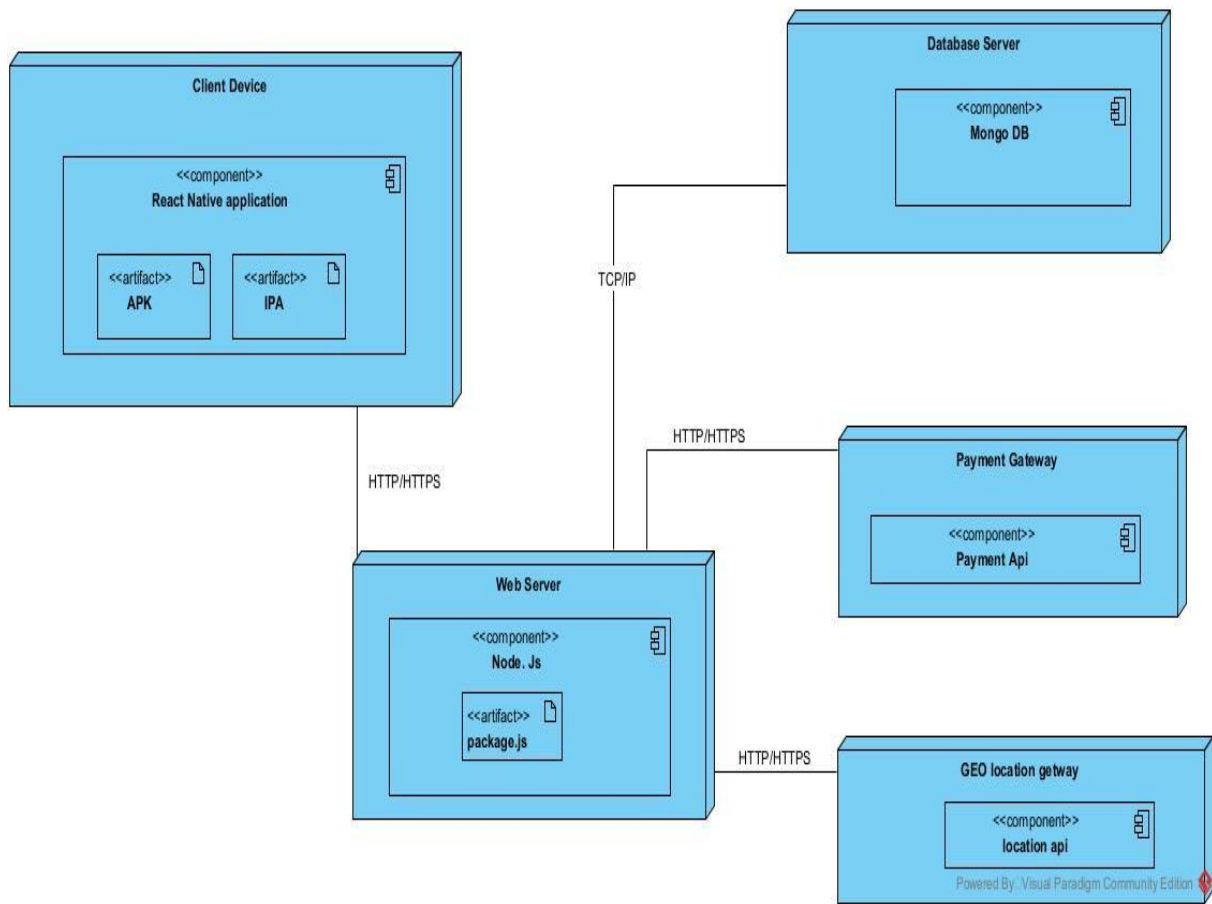


Figure 29 Pharma Connect Deployment diagram

II. COMPONENT DIAGRAM

i. DEFINITION

Component diagrams are used to model the physical aspect of a system. Now the question is, what is this physical aspect? They are elements such as Executables, libraries, files, documents, etc., which reside in a node. The component diagram does not describe the *functionality of the system*, but it describes the components used to make those functionalities.

ii. FORMALISM

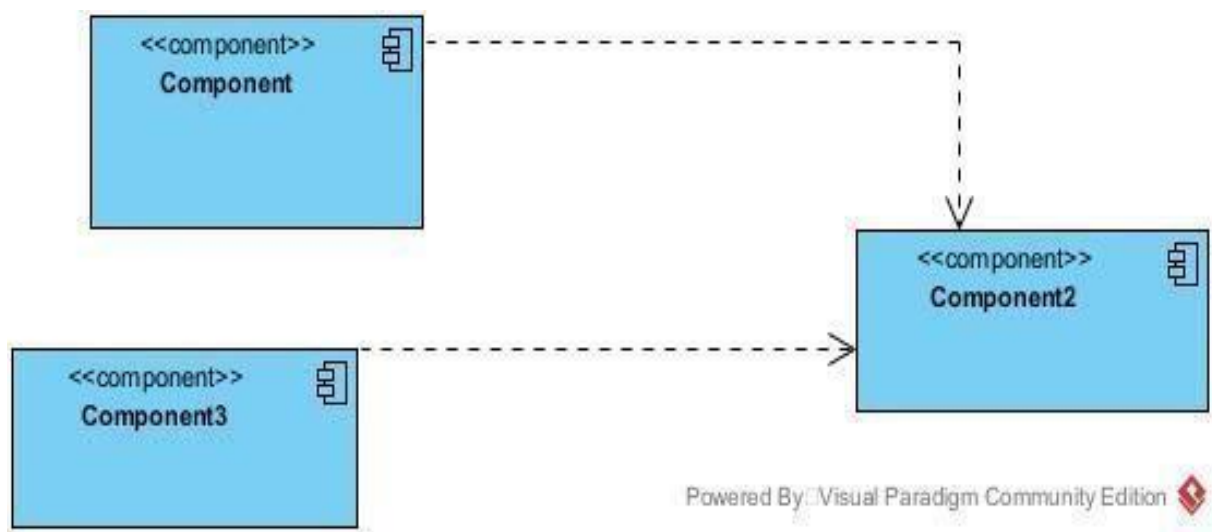
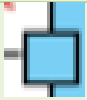
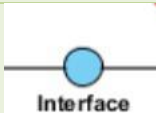
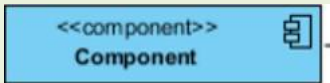


Figure 30 Formalism of Component diagram

iii. COMPONENTS OF COMPONENT DIAGRAM

Table 23 Components of component diagram

Element	Notation	Description
Dependency		A dependency is a relationship that signifies that a single or a set of model elements requires other model elements for their specification or implementation.
Port		A port is often used to help expose required and provided interfaces of a component.
Interface		It is a point of interaction between a component and its environment or other components. It specifies the services that a component provides or requires from other components
Component		It represents a self-contained and replaceable unit that interacts with other components via provided and required interfaces

iv. COMPONENT DIAGRAM FOR PHARMACONNECT

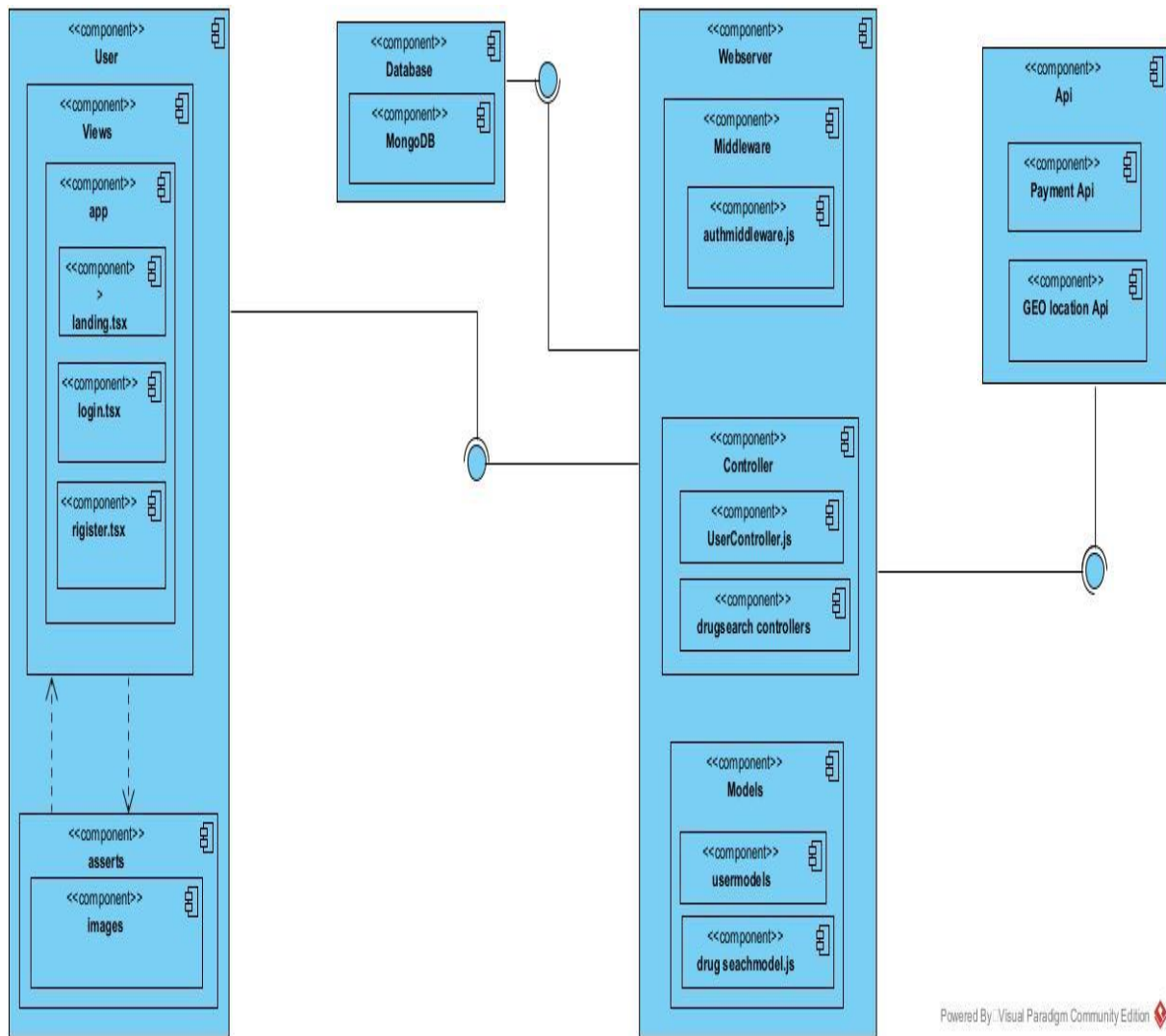


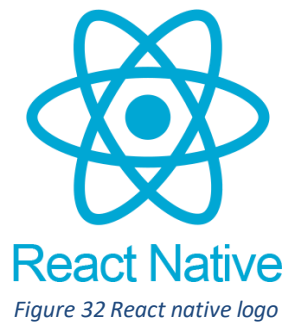
Figure 31 PharmaConnect Component diagram

III. TECHNICAL CHOICES

To set up our platform and meet the needs of the end-users, the technologies chosen are React Native for the front-end, combined with Node.js for the back-end, and MongoDB for interaction with the database. Each of these technologies will be presented in detail in the following paragraphs.

a. React Native;

React Native is an open-source framework. It allows developers to build mobile applications using JavaScript. In software development, a framework serves as a platform for developing applications. It contains resources like pre-built components, libraries, and reference material. It enables the creation of cross-platform apps that can run on both iOS and Android using a single codebase, which saves development time and effort.



b. Node js;

Node.js is an open-source, cross-platform JavaScript runtime environment that allows developers to run JavaScript code on the server side. It uses Chrome's V8 JavaScript engine to execute code and is designed to build scalable and high-performance applications, especially for web services and APIs.



Figure 33 node jess logo

c. MongoDB;

MongoDB is an open source, cross-platform, document-oriented database program. Classified as a NoSQL database product, MongoDB utilizes JSON-like documents with optional schemas. MongoDB is developed by MongoDB Inc.



Figure 34 MongoDB logo

d. Gantt Project;

Project Gantt is a widely used project management tool for planning and tracking project tasks over a given period. A Gantt chart is particularly useful for visualizing dependencies between tasks, start and finish dates, and project progress.



Figure 35 Gantt Project

e. Visual paradigm;

Visual Paradigm Community Edition is a UML software that supports all UML diagram types. It is an international award-winning UML modeler, and yet it is easy-to-use, intuitive & completely free.



Figure 36 visual paradigm logo

f. Postman

Postman is a popular API development tool that facilitates the creation, testing and documentation of APIs. It is widely used by developers, testers and API development teams to simplify the process of working with HTTP APIs.



Figure 37 postman logo

CONCLUSION

During our realization phase, we successfully developed and implemented our application. To achieve this, we utilized our analysis and conception phase. We also created deployment and component diagrams which illustrate the modules, files, and assets, as well as how different elements interact with each other within our system. Our next step is to move onto the functionalities testing phase, where we will assess the efficacy of the various modules present in our app for different users.



FILE 6 : FUNCTIONALITY TESTING

Preamble

This phase aims at testing the software or application to verify if the system is working as expected. It involves testing individual functionalities and capabilities of the software to ensure that they work correctly and meet the specified needs.

Content

INTRODUCTION

I. APPLICATION FUNCTIONALITIES

II. TEST

CONCLUSION

INTRODUCTION

It is a systematic process that involves evaluating the functionality of a software application or system in different phases or stages of its development lifecycle. This testing approach ensures that each phase of the software development process is thoroughly tested for its intended functionalities and overall performance. Its main goal is to identify and rectify any defects and bugs during the process.



I. APPLICATION FUNCTIONALITIES

- User Registration & Authentication – Users, pharmacies, and admins can create accounts and log in securely.
- Drug Search & Inquiry – Users can search for medicines and send availability requests to nearby pharmacies.
- Pharmacy Response System – Pharmacies receive drug requests and respond whether the medicine is available or not.
- Drug Availability Display – Users can view which pharmacies have the drug in stock, along with details (price, location).
- Location & Map Integration – Users can locate pharmacies on a map and get directions.
- Admin Management – Admin can verify pharmacy details, manage users/pharmacies, and oversee the platform.
- Notification System – Real-time alerts to users when pharmacies respond to drug inquiries.



II. TEST

a. DEFINITION

Testing, in simple terms, involves verifying that your code functions correctly. It is important because it boosts your confidence in the code and ensures its reliability. There are different types of testing, including:

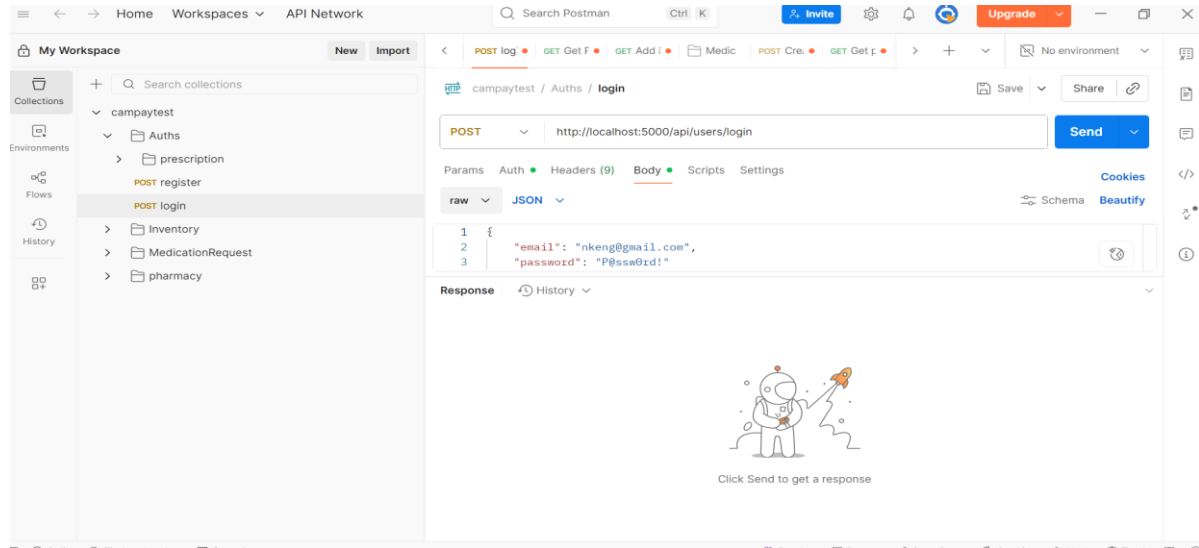
- **Unit test:** In a unit test, you are testing whether each function is working properly.
- **Feature test:** In a feature test, you are testing whether your routes, controllers, and database behave as expected.
- **Manual testing:** Here, you are manually testing your application.
- **Alternative or Automated testing:** Using online tools to test your code.

b. TESTS MADE IN THE SCOPE OF OUR PROJECT

For the test of functionality, we've used Postman. Postman is a popular API development tool that facilitates the creation, testing and documentation of APIs. It is widely used by developers, testers and API development teams to simplify the process of working with HTTP APIs.

POST MAN INTERFACE

Figure 38 Postman Interface



REGISTRATION TEST

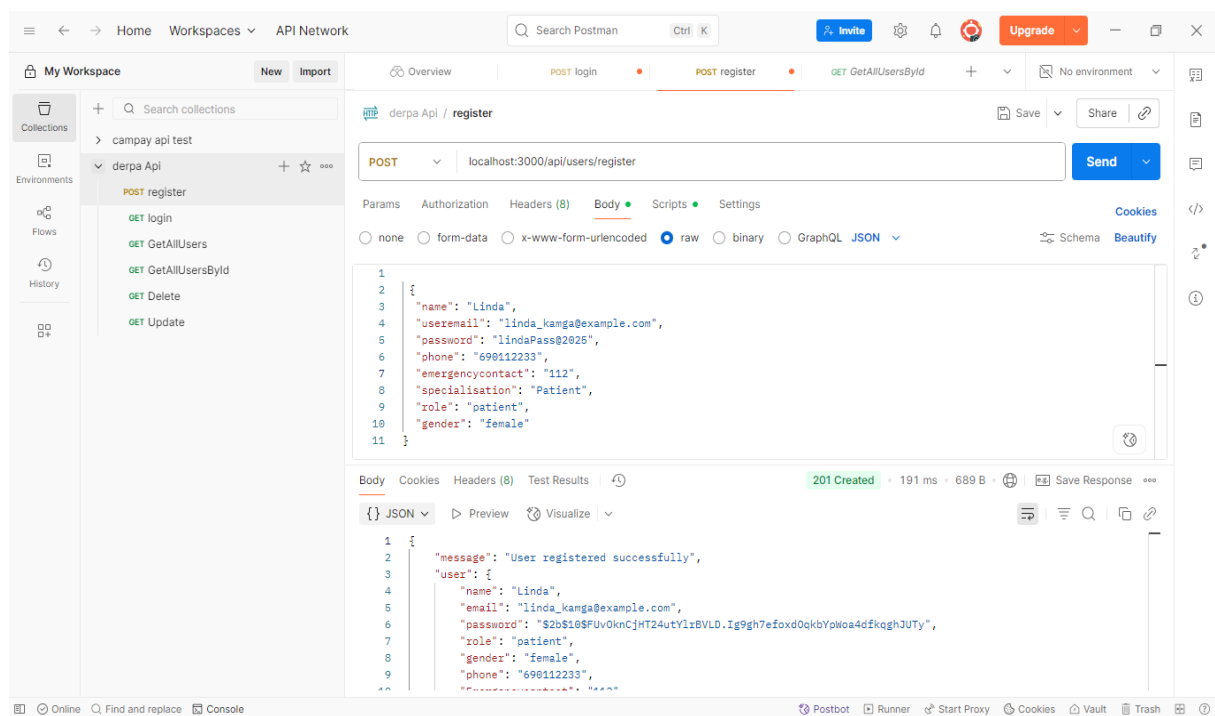


Figure 39 Registration test

LOGIN TEST

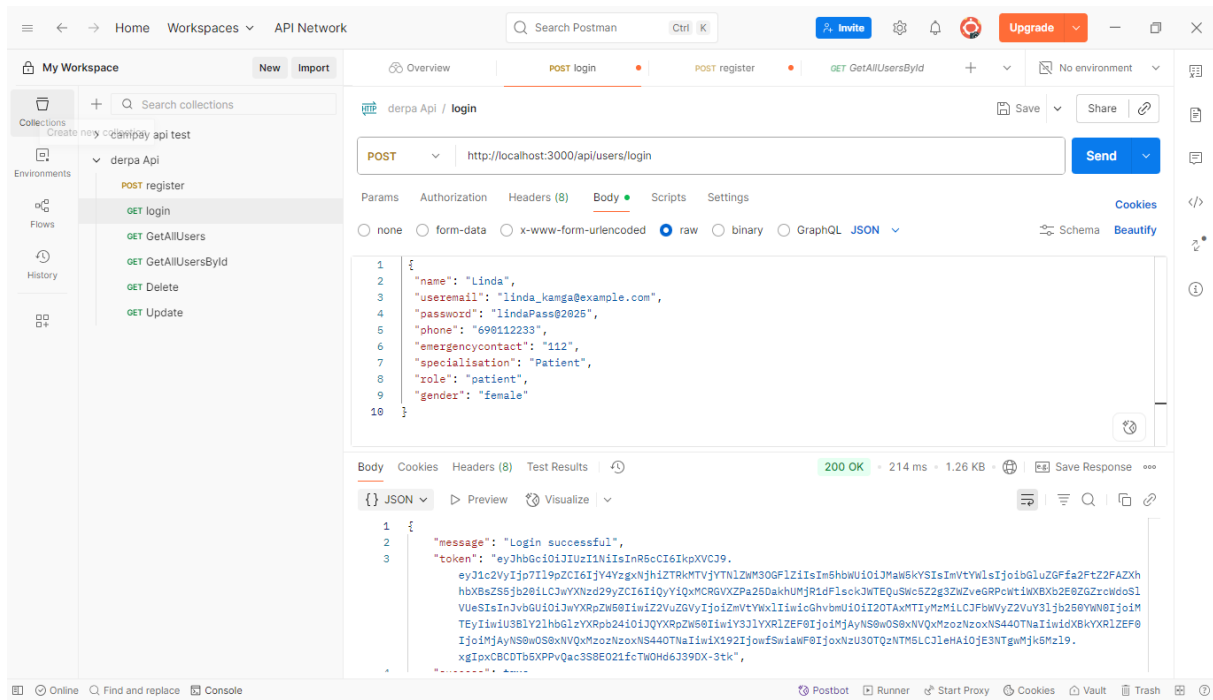


Figure 40 Login test

DELETE USER TEST

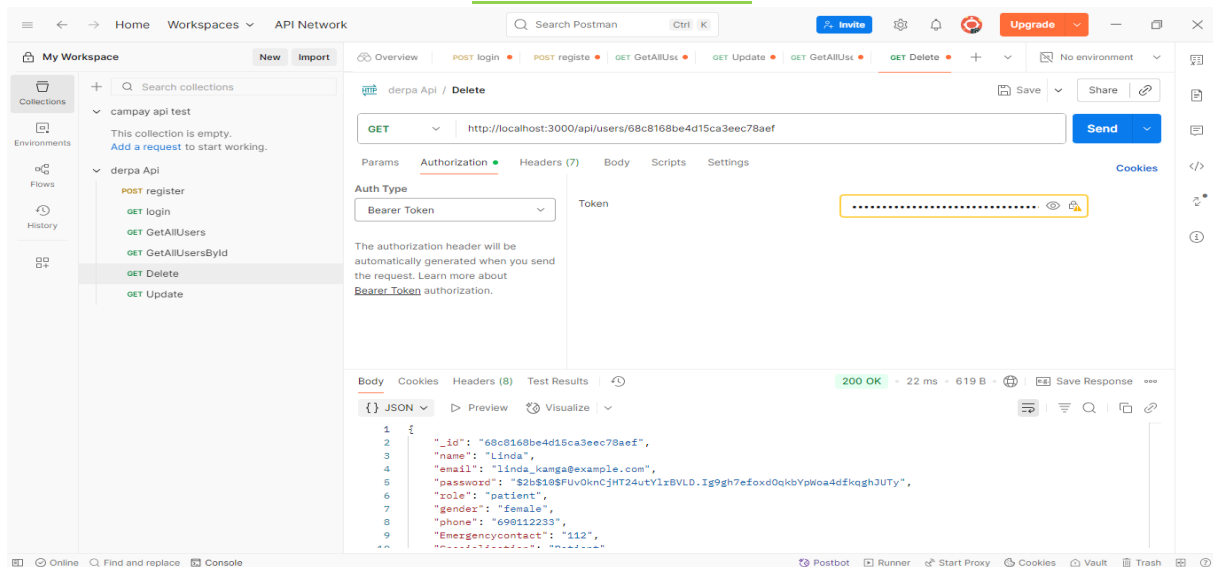


Figure 41 Delete user

CONCLUSION

Testing is an integral part of the software development life cycle and is performed to verify the correctness, reliability, and robustness of the implemented functionality. The functionality and testing phase is therefore a critical stage in software development where the implemented features are validated, and any issues are identified and resolved.

FILE 7 : USER GUIDE

Preamble

A user guide explains how to use a software application in a language that a non-technical person can understand. Thus, it enables the user to easily use the application to familiarize themselves with the software and discover all its functionalities.

Content

INTRODUCTION

I. DESCRIPTION OF THE APPLICATION

II. USER MANUAL

CONCLUSION

INTRODUCTION

The user manual is the document established after the realization of an application, software, or platform. It informs on the question "how to use the software, the application, or the platform that you have in front of you? ". It is therefore crucial for us to offer the various users of our platform a guide allowing them to easily perform various operations on our platform.

I. DESCRIPTION OF THE APPLICATION

PharmaConnect addresses the challenge of quickly locating available medications at nearby pharmacies, reducing time spent searching for drugs and improving access to essential medications.

Core Value Proposition: Connects users with nearby pharmacies in real-time, providing drug availability information and enabling fast, efficient medication procurement.

Key Features for PharmaConnect:

- **User Registration and Authentication:** Create accounts for patients and pharmacy partners, with guest access for basic searches
- **Location-Based Pharmacy Discovery:** GPS integration to find nearby registered pharmacies with distance and navigation support
- **Real-Time Drug Availability Search:** Search specific medications and view current stock levels at participating pharmacies
- **Pharmacy Profiles:** Detailed information including operating hours, contact details, specialties, and user ratings
- **Reservation System:**
- **Prescription Upload:** Upload prescriptions for pharmacy verification and preparation
- **Notification System:** Alerts for drug availability, price changes, and reservation confirmations
- **Emergency Locator:** Quick access to 24-hour pharmacies for urgent medication needs
- **User Reviews and Ratings:** Community feedback on pharmacy services and experiences

USER MANUAL

➤ PREREQUISITES

Before using PharmaConnect, ensure you have:

- An Android smartphone/ ios iPhone
- The PharmaConnect APK file downloaded.
- A stable internet connection (for real-time features such as risk).

➤ USEFUL STEPS

Below are the steps you will follow to install and use the application:

➤ **Download and Install**

Download the PharmaConnect APK file.

Open it on your phone and follow the prompts to install.

➤ **Launch the App**

Tap the PharmaConnect icon to open the application.

➤ **Landing Screen**

Choose to create an account, log in, or continue without authentication.

➤ **Sign Up**

Enter your details (name, email, password, phone, emergency contact, role, gender).

➤ **Log In**

Use your email and password to access your account.

➤ **Home Page**

View events, announcements, and awareness updates.

Drug request/ Search Test Page

Input the medication, upload image or take photo

2. SHOW CASE

- **Landing Screen:** Create an account, log in, or continue as a guest.



Figure 42 Landing Screen

- **Sign Up Screen:** Register with personal and health details.

Figure 43 sign up screen

- **Log In Screen:** Access your account securely.

Figure 44 login screen

- **Role Page:** Select user type for tailored experience.

Figure 45 role screen

- **Patient dashboard**

Figure 46 patient dashboard

- **Medication Search Page:** Perform genotype compatibility check.

Figure 47 drug search test match screen

Figure 48 chatbot screen

CONCLUSION

This user guide has provided a comprehensive overview of our product's features, functionalities, and best practices for optimal use. By following the instructions and tips outlined in this document, you'll be well-equipped to maximize the potential of our product and streamline your workflow.

PERSPECTIVES

Having developed the core functionalities of our application, we plan to improve it further with the following features:

User Perspective:

- Fast medication location and availability checking
- Cost comparison and reservation capabilities
- Personalized medication management and reminders
- Secure payment options and insurance integration
- Access to professional pharmaceutical consultations

Pharmacy Perspective:

- Increased customer reach and visibility
- Real-time inventory management integration
- Streamlined prescription processing and verification
- Revenue opportunities through consultation services
- Enhanced customer relationship management

Healthcare System Perspective:

- Improved medication adherence and patient outcomes
- Reduced prescription errors through digital verification
- Better chronic disease management and monitoring
- Enhanced coordination between pharmacies and healthcare providers
- Data-driven insights for public health initiatives

Business/Economic Perspective:

- Revenue through transaction fees and premium services
- Partnership opportunities with insurance providers and healthcare institutions
- Scalable business model with multiple monetization streams
- Market expansion potential across healthcare ecosystems
- Cost reduction in healthcare delivery through improved efficiency



Technical Perspective:

- Robust security infrastructure with blockchain integration
- Scalable cloud-based architecture for real-time data processing
- AI-powered recommendation systems and analytics
- Multi-platform compatibility and offline functionality
- Integration capabilities with existing healthcare systems



GENERAL CONCLUSION

Having come to the end of our project and our internship at REALIZE, we can proudly say that this experience has been both enriching and challenging. Throughout the three months, we were faced with moments of stress, failures, and successes, but each step helped us grow both professionally and personally. Our time at REALIZE was very beneficial as it allowed us to adapt to the professional environment, strengthen our teamwork spirit, and apply the knowledge acquired in class to real-world problems. For our project, we worked on the theme: **“Conception and Realization of an application To Connect Users with Nearby Pharmacies for Fast Drug availability.”** Our greatest desire is to contribute to reducing the early death rate caused by sickle cell disease by equipping families with accessible tools for awareness, genotype risk assessment, and preventive guidance. We began by identifying the existing challenges in the current system for sickle cell healthcare in Cameroon and formalized them in our specification book. We then carried out an in-depth analysis using UML and 2TUP methodology and modeled our system with the help of Visual Paradigm. For the realization, we made use of **React Native** for the mobile front-end, **Node.js** for the backend, and **MongoDB** as our database, ensuring a reliable and scalable architecture. The result of our work, PhamaConnect, will allow families to easily access genotype compatibility information, receive educational support, consult verified healthcare providers, and benefit from an integrated chatbot assistant for guidance. We will not stop here. In the near future, important updates and improvements will be made in terms of functionalities and security in order to make the system more reliable and impactful for communities across Cameroon and, hopefully, beyond.

ANNEXES

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CONCEPTION AND REALIZATION OF AN APPLICATION TO CONNECT USERS WITH NEARBY PHARMACIES FOR FAST DRUG AVAILABILITY



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